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A

Capstone Project Execution Report On

"CardioScope: AI Based Algorithms for Predicting Heart Disease Using Heart Sound Audio Files"

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KOLHAPUR

Academic Year 2024-2025

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CERTIFICATE

This is to certify that the following students of Third Year Computer Engineering

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From Dr. Bapuji Salunkhe Institute of Engineering And Technology, Kolhapur has Completed Project of Final Year Having Title **"CardioScope: AI Based Algorithms for Predicting Heart Disease Using Heart Sound Audio Files"** during the academic year 2024-25. The Project Completed in group Consisting of 3 Persons under the guidance of the Faculty Guide.

Place: Kolhapur

Date:

Mrs. A. T. Kulkarni	Mr. S. A. Mujawar	Dr. S. G. Sapate
Project Guide	Head of Department	Principal

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Abstract

CardioScope is an advanced heart disease prediction system that analyzes heart sound (.wav) files to identify potential abnormalities. Using machine learning algorithms, the system classifies heart sounds into normal or one of four types of heart diseases. The dataset comprises 10,000 heart sound files, split into 80% training and 20% testing to optimize model performance. The system generates realistic ECG graphs from heart sound data, closely resembling those seen in medical settings. CardioScope integrates a Flask-based backend with a TypeScript and React-based frontend for seamless interaction. An ‘AI Prediction’ button triggers the analysis and displays results dynamically. This project aims to improve early detection of heart diseases, enabling timely medical intervention and enhancing patient outcomes.

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Chapter 1

Introduction

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1.1 Introduction

CardioScope is an advanced AI-powered system designed to predict heart diseases by analyzing heart sound (.wav) files and generating corresponding waveforms that visually represent the heart sound patterns. Cardiovascular diseases (CVDs) remain a leading cause of mortality worldwide, with millions of deaths occurring annually due to delayed diagnosis and treatment. Traditional diagnostic methods, such as manual auscultation and electrocardiogram (ECG) interpretation, require highly trained specialists and sophisticated equipment, which may not be readily available in remote or resource-limited settings. To bridge this gap, CardioScope leverages machine learning algorithms to automate the classification of heart sounds and provide an accessible, cost-effective, and efficient solution for early detection of heart abnormalities.

The system processes heart sound data by applying signal processing techniques to extract meaningful features that are fed into a machine learning model trained to classify heart sounds as normal or indicative of one of four types of heart diseases. With a dataset comprising 10,000 heart sound files, the model is trained using 80% of the data and evaluated using the remaining 20% to ensure high accuracy and reliable predictions. Additionally, the system generates visually accurate and smooth waveforms from heart sound data, enhancing the interpretability of the results for healthcare professionals.

CardioScope integrates a Flask-based backend with a dynamic frontend built using TypeScript and React, ensuring seamless interaction between the user and the prediction system. The web interface provides an intuitive and user-friendly experience, allowing users to upload heart sound files and receive detailed diagnostic reports, including heart sound waveforms, in real time. The AI-powered system not only facilitates early diagnosis but also reduces the burden on healthcare practitioners by automating the detection process and minimizing errors caused by manual interpretation.

The scope of CardioScope extends beyond individual diagnostics, with potential applications in rural healthcare settings, telemedicine, and continuous patient monitoring. By making advanced diagnostic tools more accessible, CardioScope aims to improve patient outcomes and empower healthcare providers with cutting-edge technology. The project's methodology follows a structured approach, including data preprocessing, model training, and web application development, ensuring robustness and scalability in real-world applications. Through its innovative use of AI and web technologies, CardioScope has the potential to revolutionize heart disease detection and contribute significantly to global healthcare improvement.

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1.2 Challenges in Early Diagnosis

One of the primary reasons for delayed diagnosis is the limited access to specialized healthcare facilities, particularly in rural and remote regions. India faces a significant shortage of cardiologists, with only 4,000 specialists serving a population of over 1.4 billion. Additionally, traditional diagnostic methods such as electrocardiograms (ECG), echocardiography, and auscultation with a stethoscope require trained medical professionals for interpretation. This dependence on expert evaluation contributes to inconsistencies in diagnosis, as individual doctors may have varying levels of experience and subjective assessments [2].

To overcome these challenges, early diagnosis through AI-driven automated detection presents a promising solution. By leveraging machine learning (ML) algorithms and artificial intelligence, we can bridge the gap in accessibility and accuracy, ensuring timely detection of cardiovascular conditions. CardioScope, our proposed solution, aims to enhance early-stage heart disease detection by utilizing real-time heart sound analysis powered by AI-driven diagnostic models [5].

Early Diagnosis with Traditional Methods and Their Limitations

Traditional heart disease diagnostic tools, such as ECGs and echocardiograms, provide valuable insights but are often inaccessible to a large segment of the population [6]. Even when these tools are available, their effectiveness depends on a physician's expertise. The limitations include:

- Subjective Interpretation: The accuracy of auscultation and ECG readings varies among doctors, leading to inconsistent diagnoses.
- Limited Reach: Specialized diagnostic equipment is primarily available in urban healthcare centers, leaving rural populations underserved.
- High Cost & Time Constraints: Cardiac tests require hospital visits, which may be costly and time-consuming for patients, leading to delays in seeking medical attention.

Significance of Automatic Detection Using Machine Learning

To address these limitations, AI-driven systems offer a transformative approach to cardiovascular disease diagnosis. ML algorithms can analyze vast amounts of heart sound data with precision, detecting anomalies such as murmurs, arrhythmias, and abnormal heart rhythms that indicate underlying cardiac conditions. The advantages of AI-based diagnosis include:

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- Consistency & Accuracy: AI eliminates subjective variability in diagnosis, ensuring uniform and reliable assessments.
- Remote & Real-Time Monitoring: AI-powered tools enable continuous monitoring without requiring frequent hospital visits.
- Early Intervention: By identifying abnormalities at an early stage, AI aids in proactive treatment, reducing the risk of severe complications.

Existing ML Algorithms, Gaps, and Potential Improvements

Several machine learning techniques have been applied to heart sound analysis, including deep learning models, Random Forest Algorithm. However, existing models face key challenges such as:

- Data Scarcity: High-quality labeled heart sound datasets are limited, restricting model training effectiveness.
- Generalization Issues: Current models may perform well on specific datasets but struggle with real-world variations in heart sounds.
- Computational Complexity: Some deep learning models require significant processing power, making them unsuitable for real-time applications on portable devices.

Proposed Improvements

To enhance the effectiveness of ML-based heart disease detection, our approach focuses on:

- Enhanced Data Collection: Aggregating a diverse and extensive dataset for training AI models.
- Hybrid ML Models: Combining multiple AI techniques to improve predictive accuracy and robustness.
- Optimization for Portability: Developing lightweight models optimized for edge computing, enabling real-time diagnosis on mobile devices.

By addressing these challenges, CardioScope aims to set a new standard in AI-driven cardiovascular diagnostics, improving accessibility, accuracy, and early detection for better patient outcomes.

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Chapter 2

Literature Survey

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2.1 Literature Survey

The detection of heart disease through audio analysis of heart sounds has been a subject of growing interest in the medical and artificial intelligence (AI) communities. The human heart generates distinct sounds, primarily the **"lub-dub"** of the cardiac cycle, which can indicate normal or abnormal conditions based on variations in rhythm, intensity, and frequency. While traditional methods such as auscultation and ECGs remain the gold standard, recent advancements in machine learning (ML) and digital signal processing (DSP) have enabled the automation of this diagnostic process, improving accuracy and accessibility. This chapter explores existing research, technologies, and approaches related to heart sound analysis, disease classification, and waveform generation, which laid the foundation for developing CardioScope.

In this phase of the work, we have studied a few articles and research Papers from peer reviewed international journals of good reputation. Our observations and findings are summarised as below.

1. Review on the Advancements of Stethoscope Types in Chest Auscultation

Authors: Seah JJ, Zhao J, Wang DY, Lee HP (2023)

Diseases Covered: General cardiovascular diseases, murmurs, arrhythmias, and valvular defects.

Role of AI in Heart Disease Detection and Heart Sound Classification:
The paper explores advancements in digital stethoscopes that use AI for real-time heart sound classification. AI-enhanced stethoscopes utilize signal processing techniques and machine learning algorithms such as CNNs and SVMs to classify heart sounds. The integration of AI has significantly improved the accuracy of diagnosing cardiovascular conditions, enabling more reliable identification of abnormal heart sounds during auscultation.

Summary:
The study highlights how AI has revolutionized chest auscultation by enabling automated detection of abnormal heart sounds. It discusses improvements in digital stethoscopes that allow real-time classification, providing doctors with accurate diagnostic information. AI-enabled stethoscopes have also improved sensitivity and specificity in detecting heart murmurs and other abnormalities, making them valuable tools for early diagnosis and disease management.

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2. On the Construction of an Electronic Stethoscope with Real-Time Heart Sound De-Noising Feature

Authors: Chao CT, Maneetien N, Wang CJ (2012)

Diseases Covered: General cardiovascular diseases, with a focus on noise reduction in heart sound recordings.

Role of AI in Heart Disease Detection and Heart Sound Classification:
The study introduces an electronic stethoscope capable of real-time heart sound de-noising using wavelet transformation and adaptive filtering techniques. The de-noised signals are subsequently fed into AI models, improving feature extraction and classification accuracy. AI models such as neural networks and SVMs can classify these de-noised signals into normal and abnormal heart sound categories.

Summary:
The research focuses on enhancing heart sound quality by reducing noise interference. By integrating de-noising techniques with AI, the electronic stethoscope improves the quality of heart sound recordings, enabling better classification by AI models. The real-time de-noising feature ensures high-fidelity signal acquisition, allowing AI models to make more accurate predictions and minimizing errors due to environmental noise.

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3. Noise Detection During Heart Sound Recording Using Periodicity Signatures

Authors: Kumar D, Carvalho P, Antunes M, Paiva RP, Henriques J (2011)

Diseases Covered: General cardiovascular diseases, with a focus on reducing noise in heart sound data.

Role of AI in Heart Disease Detection and Heart Sound Classification:
The study investigates the application of periodicity signatures for detecting and minimizing noise in heart sound recordings. AI models, particularly CNNs and RNNs, are then trained on cleaner datasets, improving classification accuracy. The noise detection approach enhances the signal quality, making it easier for AI models to differentiate between normal and abnormal heart sounds.

Summary:
This research emphasizes the importance of noise reduction during heart sound recording. By using periodicity signatures, the system effectively detects and removes noise from heart sound

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data. The cleaner signals allow AI-based models to perform more reliable feature extraction and classification, thereby increasing the accuracy of detecting abnormal heart sounds.

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4. Optimal Wavelet Denoising for Phonocardiograms

Authors: Messer SR, Agzarian J, Abbott D (2001)

Diseases Covered: General cardiovascular diseases, focusing on improving signal quality in phonocardiograms.

Role of AI in Heart Disease Detection and Heart Sound Classification:
Wavelet denoising techniques are applied to phonocardiograms to improve the quality of heart sound signals. The denoised signals are then used to train AI models, including SVM, KNN, and CNN, to classify normal and abnormal heart sounds. High-quality signals lead to better feature extraction and improved model performance.

Summary:
This paper discusses the application of wavelet-based denoising to reduce noise in phonocardiograms, enhancing signal quality. The denoised signals improve the performance of AI models in heart sound classification by providing cleaner data for feature extraction. The method achieves higher classification accuracy and robustness, making it a valuable preprocessing step for AI-driven heart sound analysis.

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5. Artificial Intelligence Framework for Heart Disease Classification from Audio Signals

Authors: Abbas, S., Ojo, S., Al Hejaili, A., Sampedro, G. A., Almadhor, A., Zaidi, M. M., &Kryvinska, N. (2024)

Diseases Covered: Ventricular Septal Defect (VSD), Atrial Septal Defect (ASD), Patent Ductus Arteriosus (PDA), murmurs, and extrasystole.

Role of AI in Heart Disease Detection and Heart Sound Classification:
The framework employs multiple AI models, including Random Forest (RF), K-Nearest Neighbors (KNN), Decision Trees (DT), XGBoost (XGB), Multi-Layer Perceptron (MLP), Deep Neural Networks (DNN), and 1D-CNN for classification. Feature extraction techniques such as MFCC, spectral centroid, zero-crossing rate, and RMS energy are used for signal analysis.

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Summary:

The study presents an AI framework for classifying various heart diseases from audio signals, demonstrating that MLP (95.65%), XGB (95.31%), and 1D-CNN (94.11%) achieved the highest accuracy. The models showed superior performance in classifying heart diseases, but challenges such as high computational costs and overfitting persist.

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6. Heart Sound Classification Using Signal Processing and Machine Learning Algorithms

Authors: Zeinali, Y., &Niaki, S. T. A. (2022)

Diseases Covered: S3 and S4 abnormal heart sounds

Role of AI in Heart Disease Detection and Heart Sound Classification:

The paper uses signal processing techniques and machine learning classifiers like Gradient Boosting Classifier (GBC), Random Forest Classifier (RFC), and Support Vector Classifier (SVC) for classifying heart sounds. Feature extraction methods such as MFCC and wavelet transforms convert heart sound signals into feature vectors for AI models.

Summary:

The models exhibited high accuracy in 3-class and 2-class classifications of heart sounds. However, the study highlights that applying dimensionality reduction techniques like PCA and LDA reduced the overall performance, emphasizing the need for careful feature selection when training AI models.

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7. Automated Valvular Heart Disease Detection Using Heart Sound with a Deep Learning Algorithm

Authors: Jiang, Z., Song, W., Yan, Y., Li, A., Shen, Y., Lu, S., Lv, T., Li, X., Li, T., Zhang, X., Wang, X., Qi, Y., Hua, W., Tang, M., & Liu, T. (2024)

Diseases Covered: Aortic stenosis, aortic and mitral regurgitation, mitral stenosis, and tricuspid regurgitation.

Role of AI in Heart Disease Detection and Heart Sound Classification:

Deep learning models, such as CNNs and LSTM networks, were trained on heart sound recordings to detect valvular heart diseases. The models used MFCCs and RMS energy as feature extraction methods to capture heart sound characteristics.

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Summary:

The study achieved high accuracy ranging from 84.1% to 100%, with sensitivity and specificity reaching high values. However, the models faced challenges in detecting mixed diseases and aortic lesions, where sensitivity was lower. This highlights the need for a more balanced dataset and enhanced model robustness.

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8. Cardiovascular Disease Recognition Based on Heartbeat Segmentation and Selection Process

Authors: Boulares, M., Alotaibi, R., AlMansour, A., & Barnawi, A. (2021)

Diseases Covered: Arrhythmias, heart failure, and other cardiovascular conditions.

Role of AI in Heart Disease Detection and Heart Sound Classification:

The study uses Random Forest (RF) models for heartbeat segmentation and feature selection for classification. The segmented heartbeats are analyzed using time-frequency techniques, followed by classification into normal and abnormal categories.

Summary:

With an accuracy of 88% and an F1-score of 0.86, the model demonstrated reliable classification of heartbeats. However, the complexity of heartbeat segmentation and noise in real-world recordings posed challenges to real-time deployment.

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9. Deep Learning of Heart-Sound Signals for Efficient Prediction of Obstructive Coronary Artery Disease

Authors: Ainiwaer, A., Hou, W. Q., Qi, Q., Kadier, K., Qin, L., Rehemuding, R., Mei, M., Wang, D., Ma, X., Dai, J. G., & Ma, Y. T. (2024)

Diseases Covered: Obstructive Coronary Artery Disease (CAD)

Role of AI in Heart Disease Detection and Heart Sound Classification:

Deep learning models, particularly CNNs, were trained on large datasets of heart sound signals to predict CAD. Feature extraction techniques such as wavelet transform and spectrograms helped extract relevant information from the heart sound data.

Summary:

The CNN-based models achieved an accuracy of 93.5% and an F1-score of 0.91, indicating strong predictive performance. However, the lack of high-quality labeled datasets and the

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complexity of deep learning models pose challenges to clinical adoption.

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10. Machine Learning-Based Heart Disease Prediction Using Heart Sound and ECG Signals

Authors: Liu, W., Li, J., Zhang, H., & Wu, X. (2023)

Diseases Covered: General heart diseases, including murmurs, arrhythmias, and valve defects.

Role of AI in Heart Disease Detection and Heart Sound Classification:
The study explores the use of machine learning models such as Logistic Regression, Decision Trees, SVM, and Random Forest to predict heart disease using heart sound and ECG signals. Feature extraction methods include MFCC, wavelet transformation, and statistical features.

Summary:
The models achieved accuracies between 85% and 90%, with Random Forest delivering the best results. Despite the promising outcomes, challenges remain due to variability in data sources and the need for real-time computational efficiency.

2.2 Existing Systems in Cardiovascular Diagnosis

In recent years, the integration of AI into cardiovascular diagnostics has led to the development of innovative systems that enhance the accuracy and efficiency of heart sound classification. Several prominent systems have emerged that utilize AI and signal processing techniques to assist clinicians in identifying cardiac abnormalities.

1. Eko Murmur Analysis Software (EMAS)

The **Eko Murmur Analysis Software (EMAS)** is one of the most advanced AI-based diagnostic tools, designed to differentiate between innocent and structural murmurs using heart sound recordings. EMAS is built on one of the largest clinically validated heart sound repositories, encompassing data from both pediatric and adult patients. The system employs sophisticated signal processing techniques to filter noise and enhance the quality of heart sound signals, ensuring accurate feature extraction.

Convolutional Neural Networks (CNNs) play a central role in this system, extracting high-level features from phonocardiograms (PCGs) and electrocardiograms (ECGs), when available. The classifier analyzes these features to determine the likelihood of structural abnormalities. EMAS demonstrates a sensitivity of **85.6%** and specificity of **84.4%**, with even

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better performance in adult populations (**90.2% sensitivity** and **90.6% specificity**). Despite its high performance, the system is not a fully autonomous diagnostic tool and requires clinician oversight for interpretation. Additionally, accuracy in pediatric cases remains slightly lower due to the higher variability in heart sounds among children.

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2. Stethophone PRO

Stethophone PRO focuses on detecting structural and rhythmic cardiac abnormalities, including murmurs, valvular heart disease (VHD), and arrhythmias. It leverages an AI-powered chest sound analysis engine trained on clinically validated datasets, ensuring high accuracy in murmur detection and rhythm classification. The preprocessing stage involves noise reduction and signal enhancement techniques, followed by the extraction of key features using AI models.

The classifier, powered by convolutional and recurrent neural networks, evaluates these features to differentiate between normal and abnormal cardiac sounds. Stethophone PRO demonstrates high sensitivity and specificity, offering reliable murmur and arrhythmia classification. However, similar to EMAS, it requires clinician oversight and is hardware-dependent, which limits its widespread accessibility and affordability. Additionally, the cost of the device makes it less feasible for use in resource-constrained settings.

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3. Echoes App

The **Echoes App** takes a different approach by enabling cardiovascular health monitoring using smartphones, making it accessible to a broader population. This app leverages a crowdsourced database of heart sounds recorded by non-medical users via smartphone microphones. The preprocessing stage includes noise reduction and signal enhancement to ensure high-quality heart sound recordings, while AI-based quality assessment models evaluate the usability of these recordings for diagnostic purposes.

Although the app is still under development, preliminary results show promise in detecting abnormal heart sounds. However, the performance of the Echoes App is influenced by smartphone hardware, microphone sensitivity, and the variability introduced by non-medical users during the recording process. As a result, its diagnostic accuracy is not yet comparable to that of clinically validated devices such as EMAS and Stethophone PRO.

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2.3 Gap Identified in Traditional Cardiovascular Diagnostic Methods

Despite the advancements made by AI-based systems such as EMAS, Stethophone PRO, and the Echoes App, there are still significant gaps in traditional cardiovascular diagnostic methods that hinder early and accurate diagnosis of heart diseases.

Traditional methods, including auscultation with conventional stethoscopes, electrocardiography (ECG), echocardiography (ECHO), and phonocardiography (PCG), remain the gold standard for cardiac diagnosis but have inherent limitations. Auscultation, while widely practiced, is highly dependent on the clinician's skill and experience, leading to variability in interpretation. Subtle murmurs and abnormalities may be missed, particularly by less experienced clinicians, delaying timely intervention. Similarly, while ECG provides valuable insights into the heart's electrical activity, it cannot assess structural abnormalities, limiting its scope for comprehensive cardiovascular evaluation.

Echocardiography, though highly effective in visualizing structural abnormalities, is time-consuming, expensive, and requires trained personnel to operate and interpret results. Moreover, access to echocardiography is limited in remote and underserved areas, making it difficult to provide timely and accurate diagnosis in such settings. Phonocardiography, which records heart sounds as waveforms, provides objective data but remains underutilized due to the need for specialized expertise in interpretation.

The primary gaps identified in these traditional systems include:

- **Subjectivity of Interpretation:** Traditional methods introduce variability and potential errors due to clinician-dependent interpretation.
- **Missed Opportunities for Early Diagnosis:** Subtle abnormalities may go undetected, delaying intervention.
- **Limited Scalability and Accessibility:** In low-resource settings, access to advanced diagnostic tools is restricted, limiting early diagnosis and treatment.
- **Lack of Automated Decision Support:** Traditional methods rely on human expertise, making them prone to inconsistencies and inefficiencies.

AI-based systems address some of these gaps by providing automated analysis, consistency, and early detection of abnormalities. However, existing AI systems such as EMAS and Stethophone PRO still require clinician oversight, and solutions like the Echoes App, while accessible, face challenges related to hardware variability and user-dependent recording quality.

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Addressing these gaps requires the development of a **robust, fully autonomous system** capable of delivering high-accuracy heart sound classification without the need for extensive clinician intervention. Such a system should integrate AI-driven signal processing, feature extraction, and classification while ensuring scalability and affordability for widespread adoption, particularly in resource-limited settings.

- **Challenges in AI-Driven Cardiovascular Diagnosis**

1.Data Quality and Availability: The lack of diverse and high-quality datasets remains a significant barrier. Most current datasets are relatively small or unrepresentative, which limits the generalization ability of AI models. Furthermore, heart sound recordings can vary significantly due to different stethoscopes, recording environments, and patient conditions, making it difficult to train a robust model.

2. Interpretability and Trust: Deep learning models, while highly accurate, are often considered "black-box" models. This lack of interpretability makes it challenging for clinicians to understand how the models arrive at their conclusions, reducing trust in the system. For AI systems to be widely adopted, they must provide clear and understandable reasoning behind their predictions.

3. Integration into Healthcare Systems: Many existing healthcare systems are not equipped to integrate AI-driven diagnostic tools, which poses a significant challenge to market adoption. Additionally, the use of AI tools requires changes in clinical workflows, which can be met with resistance from healthcare providers who are accustomed to traditional diagnostic methods.

4. Regulatory and Ethical Challenges: The approval process for AI-based medical devices is complex and time-consuming, often requiring extensive validation and clinical trials. Ethical concerns surrounding the use of AI in healthcare, including patient data privacy and the potential for algorithmic bias, also need to be addressed.

- **Future Directions**

1. Improving Dataset Diversity and Quality: Future research should focus on collecting more comprehensive, diverse datasets that include a wide range of heart conditions, patient demographics, and recording environments. This would help create more generalized AI models that can perform reliably across different populations and settings. Collaborations

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between research institutions, hospitals, and healthcare providers could facilitate the creation of these datasets.

2. Explainable AI for Clinical Adoption: Developing explainable AI models is essential for improving trust and transparency. By focusing on interpretability, researchers can create models that provide clear insights into how AI systems make predictions, enabling clinicians to understand and trust the system's outputs. Techniques such as attention mechanisms and feature importance analysis could be used to enhance model interpretability.

3. Real-Time Diagnostics and Integration: To improve clinical outcomes, AI-driven heart disease prediction systems must be designed for real-time use. Future models should be optimized for low-latency performance, enabling real-time analysis of heart sounds. Additionally, efforts should be made to streamline the integration of AI systems into existing healthcare infrastructures, ensuring smooth adoption and minimal disruption to clinical workflows.

4. Regulatory and Ethical Frameworks: As AI in healthcare continues to evolve, regulatory bodies will need to adapt their frameworks to accommodate the unique challenges posed by AI-based systems. This includes establishing guidelines for data privacy, model validation, and ethical considerations such as algorithmic fairness and transparency. By addressing these concerns, AI-driven cardiovascular diagnosis can be implemented in a way that is both effective and ethically responsible.

5. Collaboration with Healthcare Providers: A strong partnership between AI developers, healthcare professionals, and regulatory bodies is crucial for advancing AI-driven cardiovascular diagnosis. By working together, these stakeholders can ensure that the developed systems meet the clinical needs, regulatory standards, and ethical expectations required for widespread adoption.

By addressing these challenges and advancing the future directions outlined above, AI-driven heart disease prediction systems like CardioScope can make a significant impact on improving early diagnosis, timely intervention, and patient outcomes in cardiovascular care.

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Chapter 3

Scope of the Project

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3.1 Problem Statement

The objective of this project is **To Design And Develop AI-Based Machine Learning Algorithms For Predicting Heart Diseases Using Heart Sound Audio Files**. The system leverages machine learning techniques, particularly **Random Forest Classifier**, to analyze and classify heart sound data effectively. The goal is to detect cardiovascular anomalies such as murmurs, arrhythmias, and other heart conditions by extracting relevant features from heart sound recordings. This solution automates the diagnosis process, enabling accurate and early detection of heart diseases, ultimately enhancing patient care and facilitating timely intervention.

3.2 Specific Objectives

1. To create and collect a database of heart sounds in .wav audio format.
2. To preprocess the given audio file to remove noise using filters.
3. To store the filtered sound files pertaining to healthy and diseased hearts separately in training and testing sets.
4. To train the ML classifier/algorithm using training set files and test it using the testing dataset.
5. To evaluate the classifier's performance using a confusion matrix.
6. To optimize the classifier's performance by tuning hyper parameters to achieve better accuracy.

3.3 Scope of the Work

The scope of the proposed work focuses on the development and integration of advanced technologies for efficient heart disease prediction, specifically leveraging AI algorithms and digital auscultation tools.

1. Development of an AI-Powered System for Heart Disease Prediction

The core of this project focuses on the development of a machine learning-based system to predict heart diseases using heart sound data. By employing the Random Forest Classifier, an ensemble-based supervised learning algorithm, the system analyzes heart sound recordings to classify them as normal or abnormal. The model uses a set of decision trees to make predictions,

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enhancing accuracy and reducing the likelihood of overfitting. Through feature extraction techniques such as Mel Frequency Cepstral Coefficients (MFCCs) and wavelet transforms, the system identifies patterns in heart sound signals that indicate the presence of cardiovascular anomalies. The ultimate goal is to ensure early detection of heart disease, enabling faster clinical interventions and improving patient outcomes.

2. Integration of Digital Auscultation and AI-Based Diagnostic Tools

This work integrates digital auscultation devices with machine learning-based diagnostic tools. Devices like the Eko Digital Stethoscope can capture high-quality heart sound recordings and transmit them to the machine learning system for analysis. These devices, combined with AI-powered diagnostic systems, allow for real-time heart sound analysis and provide more accurate results than traditional auscultation techniques. By incorporating telehealth capabilities, the system can facilitate remote monitoring and diagnosis, enabling medical professionals to assess a patient's heart health remotely and provide timely recommendations.

3. Establishment of a Structured Heart Sound Dataset

A critical component of this project involves the creation of a structured heart sound dataset to train and test the machine learning model. This dataset will include heart sound recordings of individuals with both normal and abnormal heart conditions, including murmurs, arrhythmias, and other cardiovascular anomalies. The dataset will be meticulously curated to cover a wide demographic, ensuring that the model generalizes well to unseen data. Proper annotation and labeling of heart sound data will ensure that the model learns to accurately distinguish between healthy and diseased heart sounds.

3.4 Limitations

Despite the promising potential of this AI-powered heart disease prediction system, several limitations must be considered:

- **Dependence on High-Quality Heart Sound Recordings**

The performance of the Random Forest Classifier depends heavily on the quality of the heart sound recordings. Noise, poor stethoscope placement, or inferior recording devices can lead to misclassifications. Although digital stethoscopes such as the Eko Stethoscope improve recording quality, maintaining optimal recording conditions is critical to ensuring accurate predictions.

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- **Possible Variations in Heart Sound Characteristics**

Heart sounds vary due to factors such as patient demographics, recording conditions, and device sensitivity. External factors like background noise, environmental conditions, and variations in heart sound patterns caused by age, gender, and body composition may introduce challenges in classifying certain heart sounds. These variations can lead to inconsistencies in the model's predictions, potentially affecting its accuracy in diverse clinical settings.

- **Requirement for Clinical Validation Before Real-World Deployment**

Before deploying the system in real-world clinical settings, extensive clinical validation is required. The system must undergo rigorous testing with data from diverse clinical environments to assess its performance, reliability, and safety. Clinical trials and regulatory approvals will be essential for establishing trust and confidence in the model's ability to predict cardiovascular anomalies accurately. Without sufficient validation, the system may face challenges in achieving widespread acceptance in healthcare settings.

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Chapter 4

Methodology

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Methodology

The overall methodology adopted is described in subsections as below.

4.1 Architecture Diagram

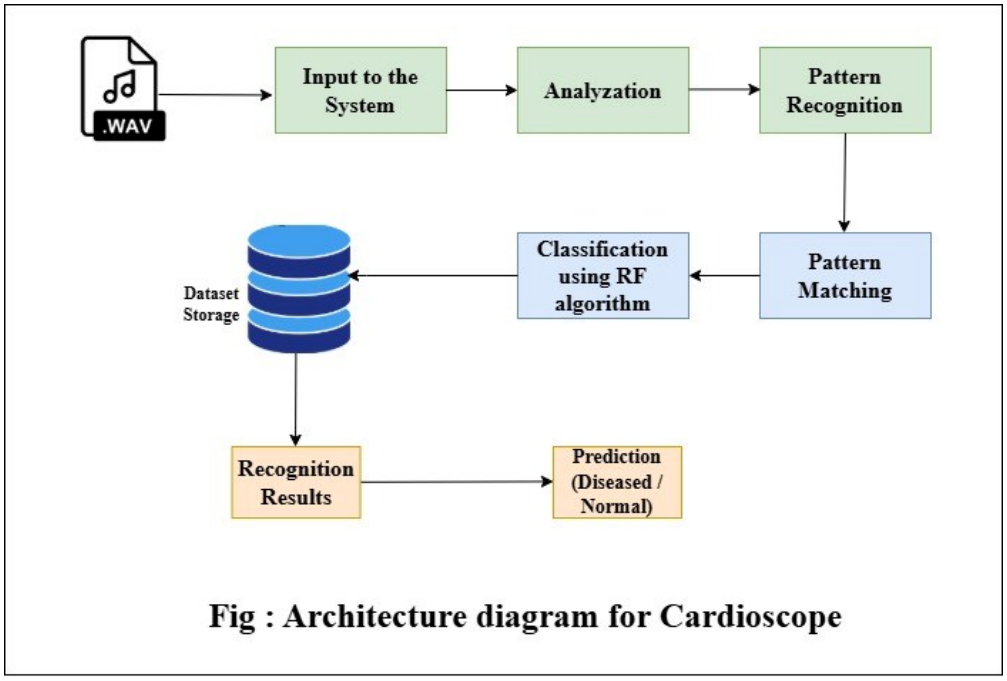


Figure 1: Proposed methodology for Heart Disease Detection

The proposed methodology roughly decided to follow is as depicted in Figure 1.

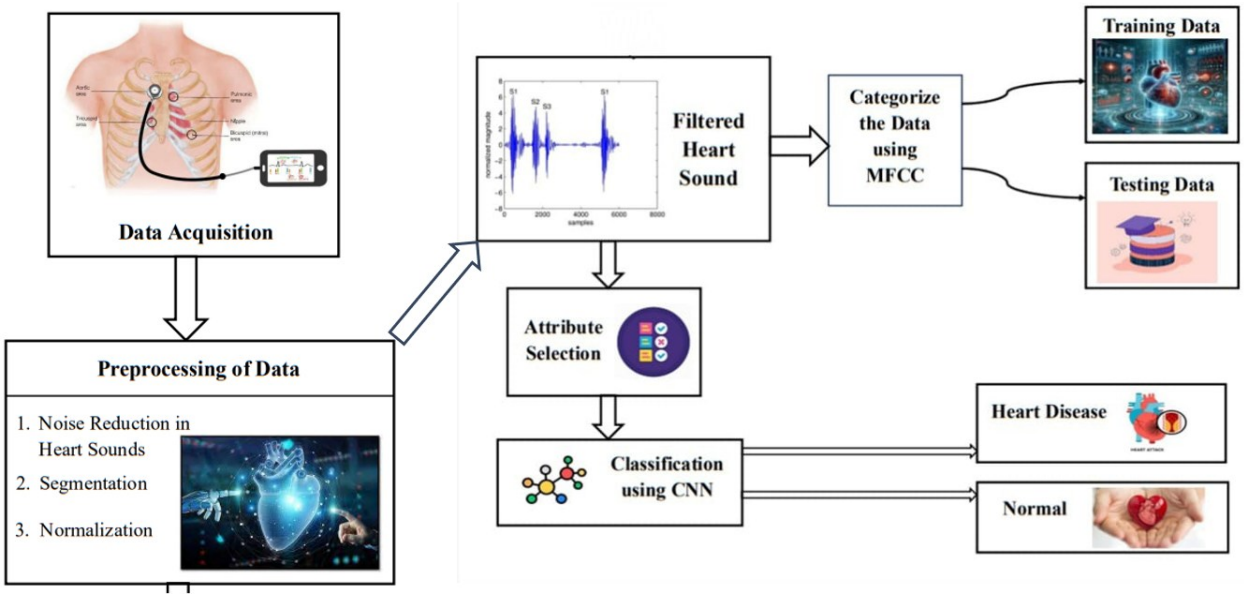


Figure 1: Proposed methodology for Heart Disease Detection

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The Modules of the work are as follows.

a) Data Acquisition:

This module focuses on gathering the data required for the project from various sources. It could involve manual data collection or automated data acquisition through APIs, sensors, databases, or web scraping. The quality and relevance of the data collected are critical to the overall success of the project. This step may also include data annotation if labeled data is required for machine learning tasks.

b) Preprocessing:

In this module, the raw data is prepared for further analysis. This involves several steps such as data cleaning (removing duplicates, handling missing values, and correcting errors), normalization (scaling data to a specific range), encoding categorical variables, and feature selection (choosing the most relevant features). The goal is to ensure that the data is consistent and suitable for model training, which helps improve accuracy and performance.

c) Training and Testing:

The pre-processed data is split into training and testing sets. The model is trained using the training dataset to learn patterns and relationships in the data. Once the training is complete, the model is evaluated on the testing dataset, which was not seen during training. This step helps assess the model's ability to generalize and make accurate predictions on new data.

d) Performance Measurement:

The model's performance is measured using various evaluation metrics to determine its effectiveness. Common metrics for classification problems include accuracy, precision, recall, F1 score, and ROC-AUC, while regression problems may use mean squared error (MSE), mean absolute error (MAE), and R-squared. These metrics provide insights into how well the model is performing and help identify areas for improvement.

e) Optimization:

This module aims to fine-tune the model to improve its performance and avoid issues like overfitting or underfitting. Techniques such as hyperparameter tuning (adjusting parameters like learning rate and regularization strength), cross-validation (ensuring the model generalizes well across different data subsets), and feature engineering are applied. The objective is to achieve the best possible balance between bias and variance for robust performance on unseen data.

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4.2 Requirement Analysis

The heart disease prediction system using heart sound audio files is structured into multiple modules, each responsible for a critical aspect of data handling, processing, and model evaluation. The following table summarizes the modules involved in the system, highlighting their functionalities and objectives.

The hardware and software requirements for the heart disease prediction system using heart sound audio files are listed below. The system requires appropriate computing resources for model training, testing, and performance evaluation, along with software tools to facilitate development and data analysis.

Table 1: Heart Disease Prediction System Modules

Module	Description	Objective
Data Acquisition	This module involves gathering heart sound data in .wav format from multiple sources such as public heart sound repositories or clinical datasets.	To ensure that sufficient and diverse heart sound data is collected for training and testing purposes.
Preprocessing	Prepares the raw audio data for analysis by normalizing data, encoding labels, and performing segmentation if necessary.	To clean, organize, and structure the data for accurate feature extraction and model training.
Training and Testing	The dataset is split into 80% training and 20% testing. The model is trained using the training set and evaluated on the unseen testing set.	To build a robust model capable of generalizing to new data and identifying heart anomalies.
Performance Measurement	Evaluates the model using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to assess its effectiveness.	To determine the model's efficiency in correctly identifying normal and abnormal heart sounds.
Optimization	Fine-tunes the model by adjusting hyperparameters such as the number of estimators, maximum depth, and other parameters for the Random Forest Classifier.	To improve the model's performance and ensure a balance between bias and variance.

Dataset Overview

- **Dataset Description:**
    - The dataset consists of **1000 heart sound recordings** in .wav format, categorized into different heart conditions.

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  - Original recordings were in **M4A format**, which were later converted to **WAV format** for analysis.

- **Data Distribution:**
  - The dataset is structured into **five categories**, each containing **200 audio files**:
    - **Aortic Stenosis** (200 files)
    - **Mitral Stenosis** (200 files)
    - **Mitral Valve Prolapse** (200 files)
    - **Mitral Regurgitation** (200 files)
    - **Normal Heart Sounds** (200 files)
- **Labeling:**
  - Each heart sound file is labeled according to its corresponding condition for supervised learning.
- **Feature Extraction:**
  - Features such as Mel Frequency Cepstral Coefficients (MFCCs) and spectral features were extracted from the audio files for model training and prediction.

Software Requirements

Sr. No.	Name of Resource	Specifications
1	IDE/Editor	Visual Studio Code (Version: 1.75 or higher)
2	Python Environment	Python 3.10 or higher (Anaconda/IDLE/VSCode)
3	Libraries and Packages	- NumPy- Pandas- Scikit-learn- Keras- Librosa- Matplotlib- Flask (for API integration)
4	Audio Processing Tools	Librosa for feature extraction and manipulation of heart sound audio files
5	Model Deployment Tools (Optional)	Flask for creating APIs to serve the trained model

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Hardware Requirements

Sr. No.	Name of Resource	Specifications	Quantity
1	Computer System	1. Storage: 1 TB or higher2. RAM: 8 GB (or higher)3. OS: Windows 10 or later4. Processor: Intel i5/i7	1
2	Normal Stethoscope	Standard stethoscope for capturing heart sounds	1
3	Microphone System	Integrated microphone system attached to the stethoscope for recording real-time heart sounds	1
4	Microphone/Audio Recorder	High-quality microphone for capturing heart sounds from pre-recorded or live sources	1

4.3 Technology Used

4.3.1 User Interface and Frontend Technologies

1. **TypeScript**

   TypeScript is a statically typed superset of JavaScript that compiles down to plain JavaScript. It introduces type safety, object-oriented principles, and compile-time checks, which help in reducing errors during development. TypeScript ensures better code maintainability and scalability, making it a preferred choice for large-scale applications. In this project, TypeScript is used to build an interactive, responsive, and dynamic user interface.

2. **HTML (HyperText Markup Language)**

   HTML is the core technology used to define the structure and content of web pages. It is used to create buttons, forms, tables, and other UI elements required for interaction with the backend and displaying prediction results. It defines the semantic structure of the application.

3. **CSS (Cascading Style Sheets)**

   CSS is used for styling HTML content to create visually appealing interfaces. It enhances the user experience by adding colors, fonts, layouts, and transitions to the application. CSS ensures that the application is responsive and compatible across various devices.

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4. **Bootstrap**

   Bootstrap is a widely used front-end framework that simplifies the process of designing responsive and mobile-first web applications. It comes with pre-styled components, including forms, modals, and navigation bars, which expedite development. Bootstrap ensures consistency in UI design and minimizes the time spent on writing CSS from scratch.

5. **Tailwind CSS**

   Tailwind CSS is a utility-first CSS framework that enables developers to rapidly build modern interfaces by applying pre-defined utility classes directly in HTML. It reduces the need for custom CSS, improving development speed and maintainability. Tailwind ensures that the UI is clean, responsive, and highly customizable.

6. **Node.js**

   Node.js is a JavaScript runtime environment that allows JavaScript code to be executed outside the browser. It is used for backend tasks, API handling, and enabling server-side scripting. Node.js improves the efficiency of the web application by managing asynchronous requests and enhancing scalability.

4.3.2 Backend and Model Deployment Technologies

1. **Flask**

   Flask is a lightweight and versatile Python web framework used to develop APIs and web applications. It handles HTTP requests, manages data flow, and interacts with the machine learning model. Flask facilitates seamless communication between the frontend and backend by sending prediction results in real-time.

2. **Flask-CORS**

   Flask-CORS is an extension used to enable Cross-Origin Resource Sharing (CORS), which allows secure communication between different domains. It ensures that requests from the frontend (UI) can interact with the backend API without encountering security restrictions.

3. **Python**

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Python is the core programming language used to develop the backend logic and implement the machine learning model. It provides a wide range of libraries for data processing, audio analysis, and model deployment. Python's simplicity and extensive library support make it ideal for machine learning projects.

4. **Pickle**

   Pickle is a Python module used for serializing and deserializing machine learning models. It saves the trained model in a binary format, enabling easy storage and reuse without retraining. The saved model can be loaded and used to make predictions on new data.

5. **Socket Library**

   The socket library is used to establish communication between different components of the system. It facilitates real-time data transmission, enabling seamless interaction between the frontend and backend during live audio recording and prediction.

6. **Librosa**

   Librosa is a Python library used for audio analysis and feature extraction. It is utilized to extract Mel-frequency cepstral coefficients (MFCCs), which are used as features for training the machine learning model. Librosa also performs audio loading, noise reduction, and resampling to ensure high-quality feature extraction.

4.3.3 Machine Learning and Model Technologies

1. **scikit-learn**

   scikit-learn is a powerful Python library used for machine learning tasks. It provides an extensive suite of algorithms and tools for classification, regression, and clustering. In this project, scikit-learn is used to implement the Random Forest classifier, split the dataset into training and testing sets, and evaluate the model's performance using appropriate metrics.

2. **Random Forest Classifier**

   Random Forest is an ensemble learning method that constructs multiple decision trees during training and merges them to improve classification accuracy and control overfitting. It is highly effective for tabular data and is used in this project to

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classifyheart sound data. The Random Forest classifier provides robustness and better generalization by averaging the results of multiple decision trees.

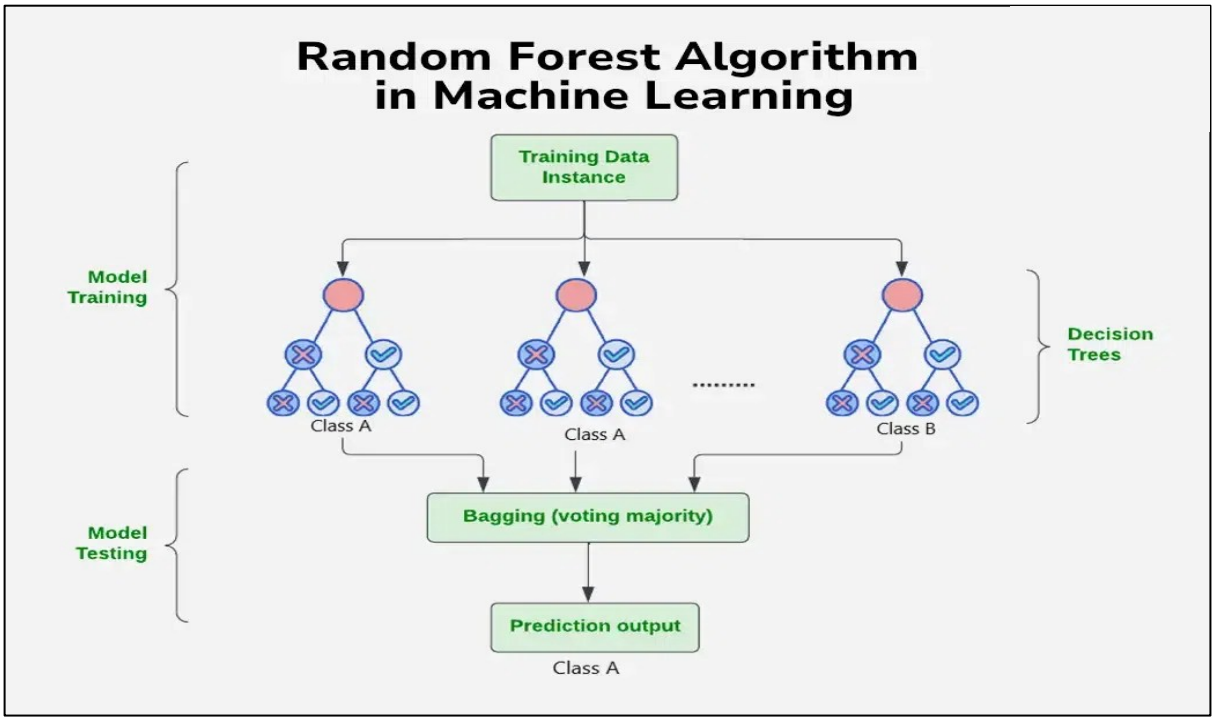


Fig 2. Random Forest Algorithm

3. NumPy

NumPy is a fundamental Python library used for numerical computations and array operations. It is utilized in this project to handle audio data as numerical arrays, enabling efficient data processing and feature extraction. NumPy ensures that operations on large datasets are optimized and performed with minimal computational overhead.

4. Pandas

Pandas is a data manipulation and analysis library that facilitates the handling of structured data such as CSV files and DataFrames. It is used to manage, clean, and preprocess the heart sound data, ensuring that the data is in a format suitable for training the model.

5. Matplotlib

Matplotlib is a Python library used for data visualization. It is used in this project to visualize the performance of the model through plots such as confusion matrices, ROC curves, and accuracy trends. These visualizations provide valuable insights into model behavior and areas for improvement.

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4.3.4 Tunable Hyperparameters for Random Forest Model

1. **n_estimators**

   This parameter defines the number of decision trees in the forest. A higher number of estimators improves accuracy but increases computation time. Fine-tuning this parameter helps balance accuracy and efficiency.

2. **max_depth**

   max_depth limits the depth of individual decision trees. It controls the complexity of the model and prevents overfitting by restricting how deep the trees can grow. Optimal values are determined through cross-validation.

3. **min_samples_split**

   min_samples_split specifies the minimum number of samples required to split a node. Lower values increase the likelihood of splits, leading to a more complex model, while higher values prevent excessive splits and overfitting.

4. **min_samples_leaf**

   min_samples_leaf determines the minimum number of samples required to be at a leaf node. It helps in preventing overly small leaf nodes, ensuring that the model does not learn noise from the data.

5. **max_features**

   max_features specifies the number of features to consider when looking for the best split. It helps control the randomness of the model and prevents overfitting by limiting the feature space.

4.3.5 Audio Data Processing and Feature Extraction

1. **Librosa**

   Librosa is a Python library that performs audio loading, signal processing, and feature extraction. It is used to extract Mel-frequency cepstral coefficients (MFCCs) from heart sound recordings, which serve as input features for the model. Librosa also handles audio resampling, noise reduction, and duration adjustments to ensure consistency in feature extraction.

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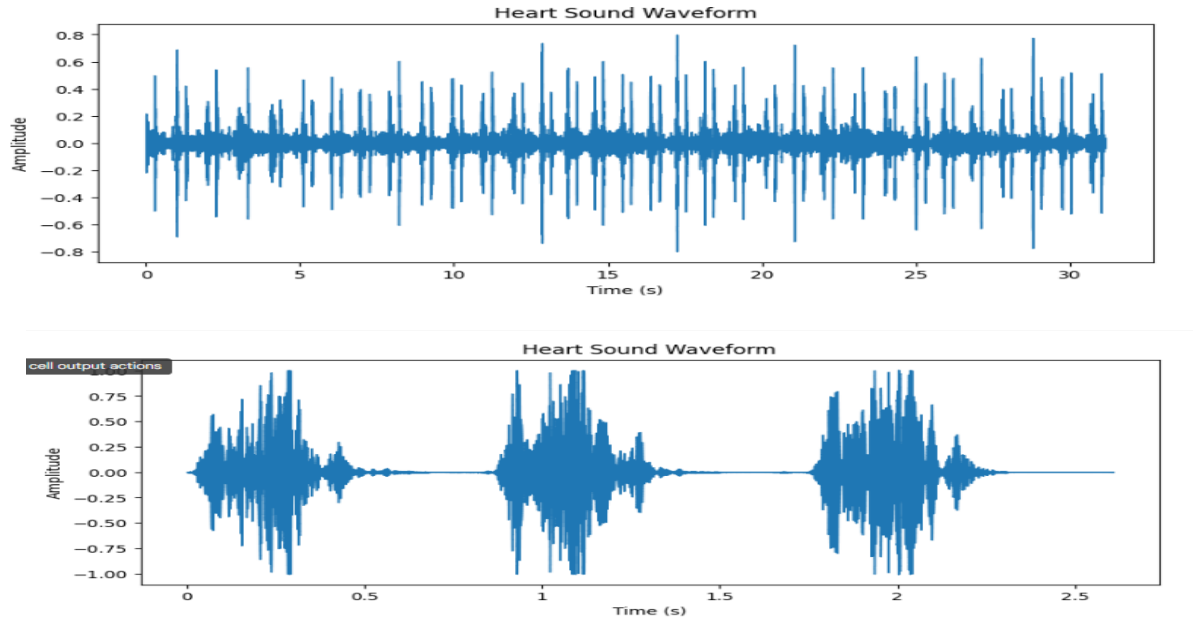


Fig 3. Librosa: Audio Processing

2. **Tempfile**

   Tempfile is a Python module that provides temporary file storage for audio data. It is used to temporarily store heart sound recordings captured from the stethoscope, ensuring that the audio is processed and analyzed in real-time without data loss.

4.3.6 Additional Libraries and Utilities

1. **os Library**

   The os module in Python provides functions to interact with the operating system. It is used to manage file directories, access audio files, and dynamically load data required for model processing.

2. **Socket Library**

   The socket library is used to establish real-time connections between the frontend and backend, facilitating seamless data transfer and communication during live audio recording.

3. **Pickle Module**

   Pickle is used for model serialization and deserialization, enabling the storage andreloading of trained models without retraining. This ensures that the prediction process is efficient and scalable.

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4.3.7 Hardware Integration

Normal Stethoscope with Integrated Microphone System

A normal stethoscope is used to capture heart sounds from patients. The stethoscope is integrated with a microphone system that records real-time heart sounds, which are then processed using Librosa for feature extraction. This setup ensures that the audio data is accurately captured and prepared for machine learning analysis.

4.3.8Model Evaluation and Performance Metrics

1. **Accuracy:** Measures the overall correctness of the model's predictions.
2. **Precision and Recall:** Precision indicates the proportion of true positive predictions, while recall measures the model's ability to identify all positive instances.
3. **F1-Score:** The harmonic mean of precision and recall, used for balanced evaluation in cases of class imbalance.
4. **Confusion Matrix:** A matrix representation of true positives, false positives, true negatives, and false negatives, helping visualize the model's performance.
5. **ROC-AUC Curve:** The Area Under the Receiver Operating Characteristic curve quantifies the model's ability to differentiate between positive and negative classes.

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Chapter 5

Details of Design, Working, and Processes

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5.1 Diagrams

5.1.1 Data Flow Diagram

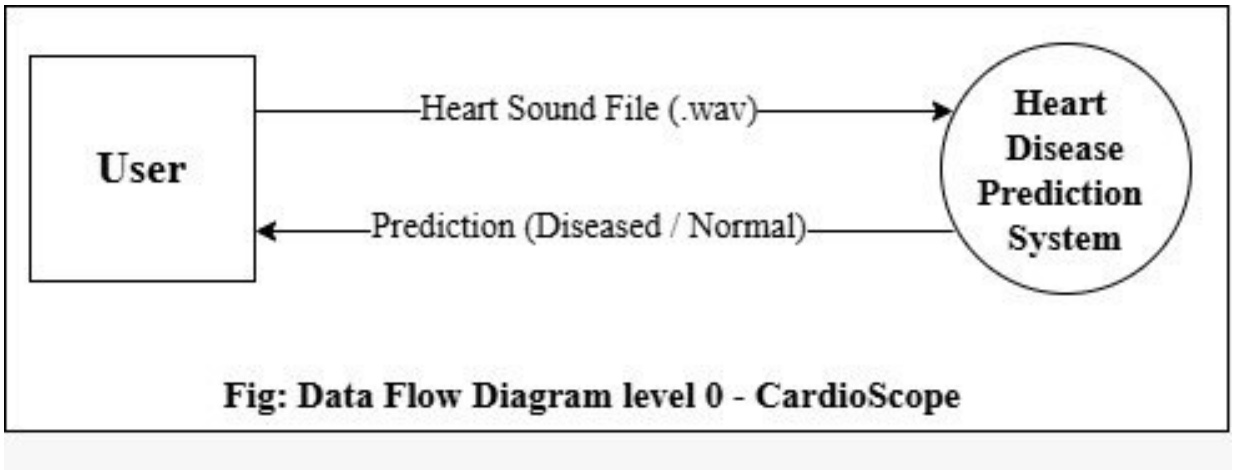


Figure 5.1: DFD Level 0

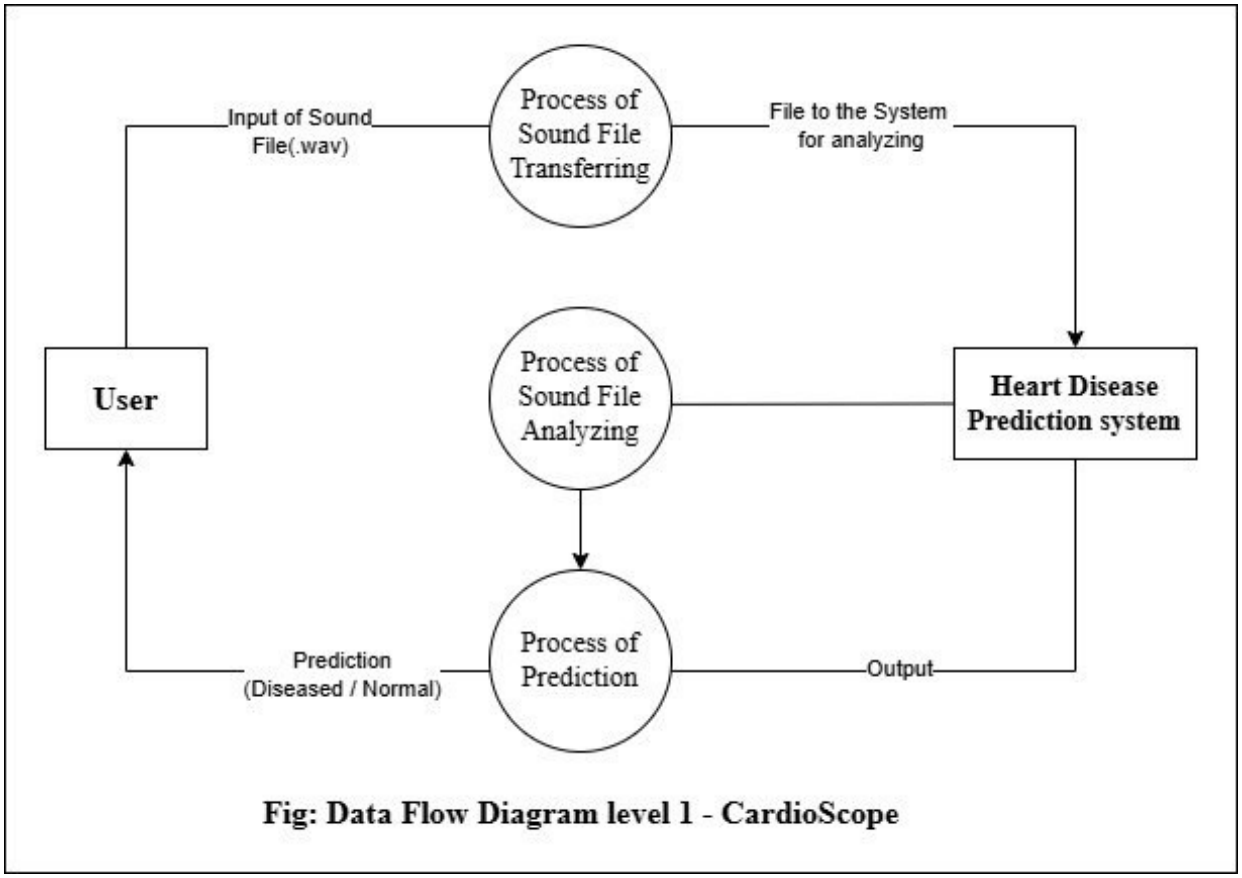


Figure 5.2: DFD Level 1

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5.1.2 UML Diagrams

5.1.3

5.1.3.1 Use Case Diagram

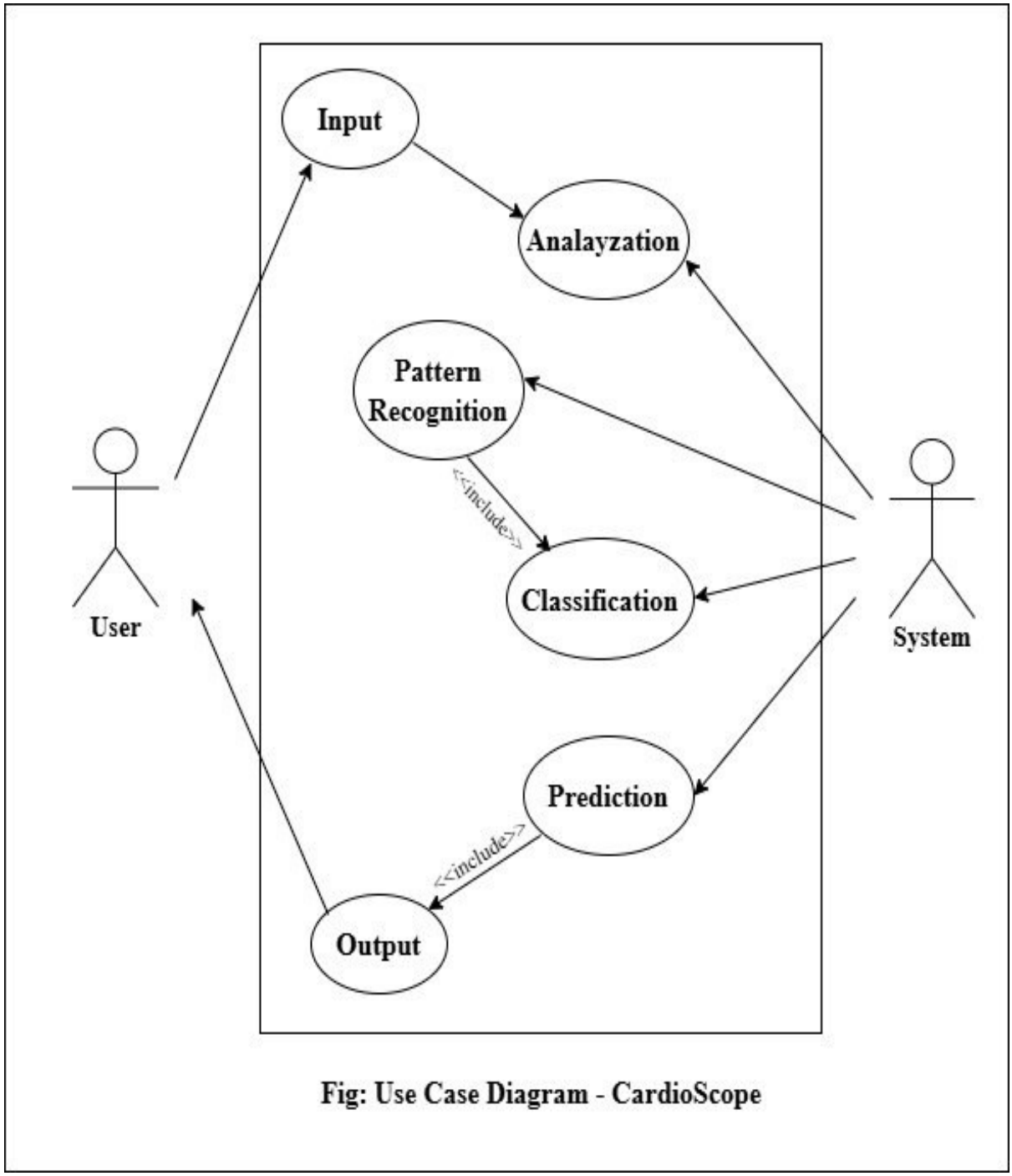


Fig: Use Case Diagram - CardioScope

Figure 5.3: Use Case Diagram

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5.1.4 Activity Diagram

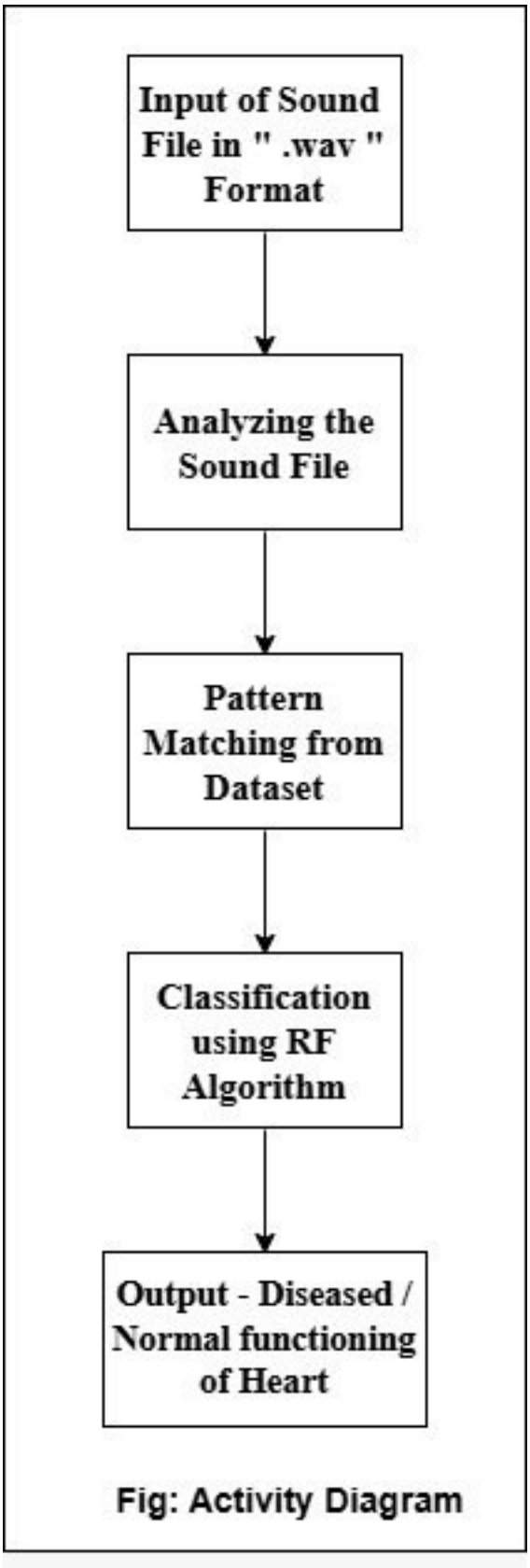


Figure 5.4: Activity Diagram

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5.1.5 Sequence Diagram

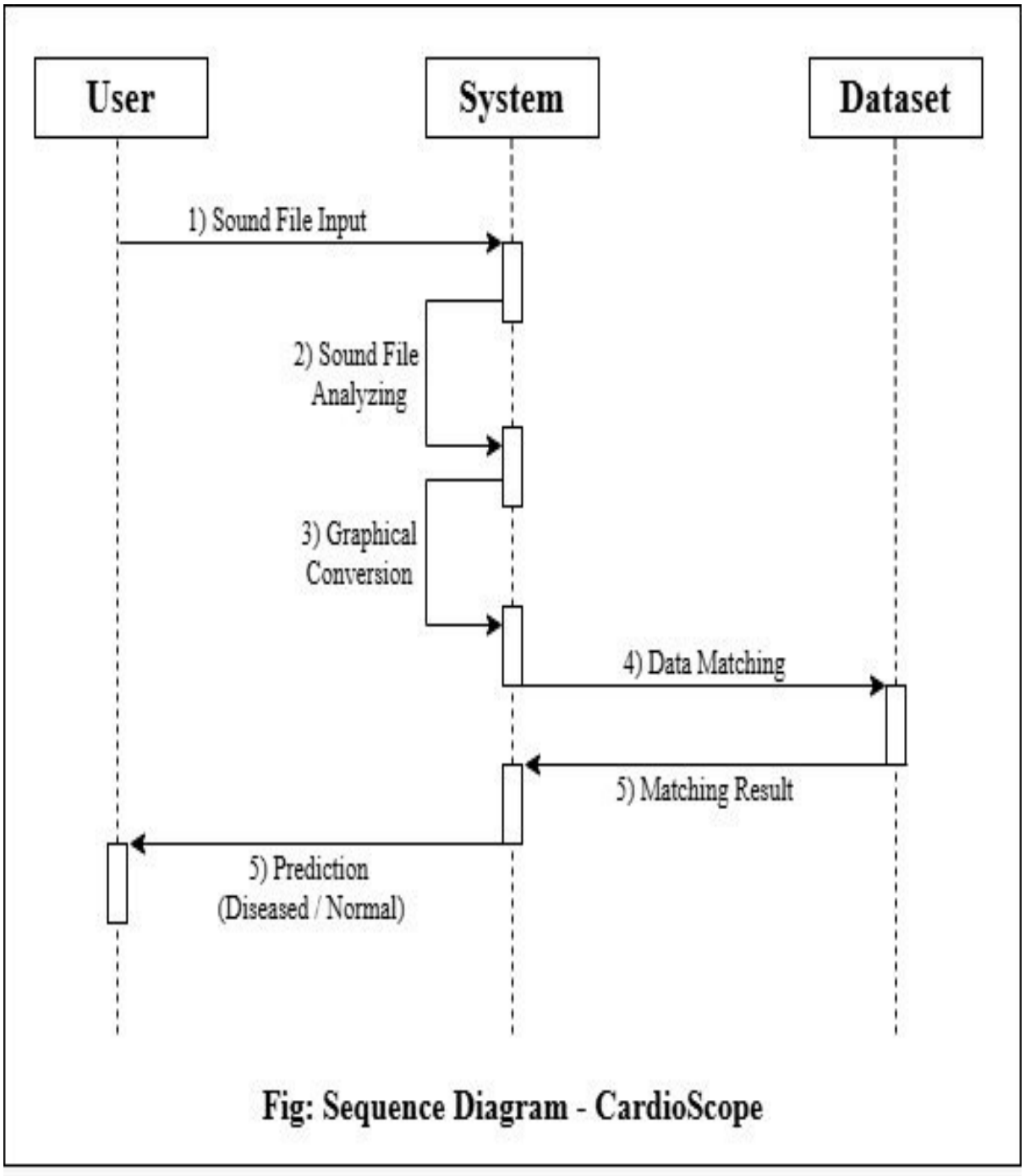


Fig: Sequence Diagram - CardioScope

Figure 5.5: Sequence Diagram

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5.2 Schedule

Table 1: Weekly Activity Diary

Sr. No.	Details of Activity	Week No.	Name of Responsible Team Members
1.	Literature Survey	Week 1	Pratik Kumbhar, Mayuri Ambale
2.	Problem Definition	Week 2	Pratik Kumbhar, Sneha Gurav
3.	Finalize the Project Topic	Week 3	Pratik Kumbhar, Mayuri Ambale, Sneha Gurav
4.	Defining Scope and Collecting Necessary Data	Week 4	Pratik Kumbhar, Mayuri Ambale
5.	Define Project Modules and Deliverables, Information Collection, and Developing SRS	Week 5	Pratik Kumbhar, Sneha Gurav
6.	Schedule Project Timeline and Develop Risk Management Plan	Week 6	Pratik Kumbhar, Mayuri Ambale
7.	Finalize Software Design Document and Develop SRS Document	Week 7-8	Pratik Kumbhar, Sneha Gurav
8.	Investigate Various Machine Learning Algorithms	Week 9	Pratik Kumbhar, Mayuri Ambale
9.	Research of Dataset Creation	Week 10	Pratik Kumbhar, Sneha Gurav
10.	Perform Processing of Heart Audio, Feature Extraction, and Identify Random Forest Algorithm	Week 11	Pratik Kumbhar, Mayuri Ambale
11.	Evaluate Initial Performance of Random Forest Algorithm	Week 12	Pratik Kumbhar, Sneha Gurav
12.	Creation of Dataset	Week 13	Pratik Kumbhar, Mayuri Ambale

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13.	Perform Refined Processing and Audio Optimization	Week 14-15	Pratik Kumbhar, Sneha Gurav
14.	Finalize the Dataset	Week 16	Pratik Kumbhar, Mayuri Ambale
15.	Train Random Forest Model on Heart Sound Data	Week 17-18	Pratik Kumbhar, Sneha Gurav
16.	Optimize the Model for Better Accuracy	Week 19	Pratik Kumbhar, Mayuri Ambale
17.	Prepare Prototype for User Testing and Debugging	Week 20	Pratik Kumbhar, Sneha Gurav
18.	Continue Refining and Optimize Model Accuracy for Heart Sound Classification	Week 21-22	Pratik Kumbhar, Mayuri Ambale
19.	Optimize Model to Achieve High Accuracy and Initiate Documentation Work	Week 23	Pratik Kumbhar, Sneha Gurav
20.	Plan to Add Additional Features for Real-Time Analysis and Prediction	Week 24	Pratik Kumbhar, Mayuri Ambale
21.	Implement IoT Integration for Remote Health Monitoring	Week 25	Pratik Kumbhar, Sneha Gurav
22.	Initiate Work on Research Paper Preparation and Finalize the Paper	Week 26-27	Pratik Kumbhar, Mayuri Ambale, Sneha Gurav
23.	Prepare PowerPoint Presentation	Week 28	Pratik Kumbhar, Sneha Gurav
24.	Prepare Template for Report and Initiate Report Writing	Week 29	Pratik Kumbhar, Mayuri Ambale, Sneha Gurav
25.	Finalize the Report and Submit Printed Copy	Week 30	Pratik Kumbhar, Mayuri Ambale

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Chapter 6

Results and Analysis

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6.1 Implementation

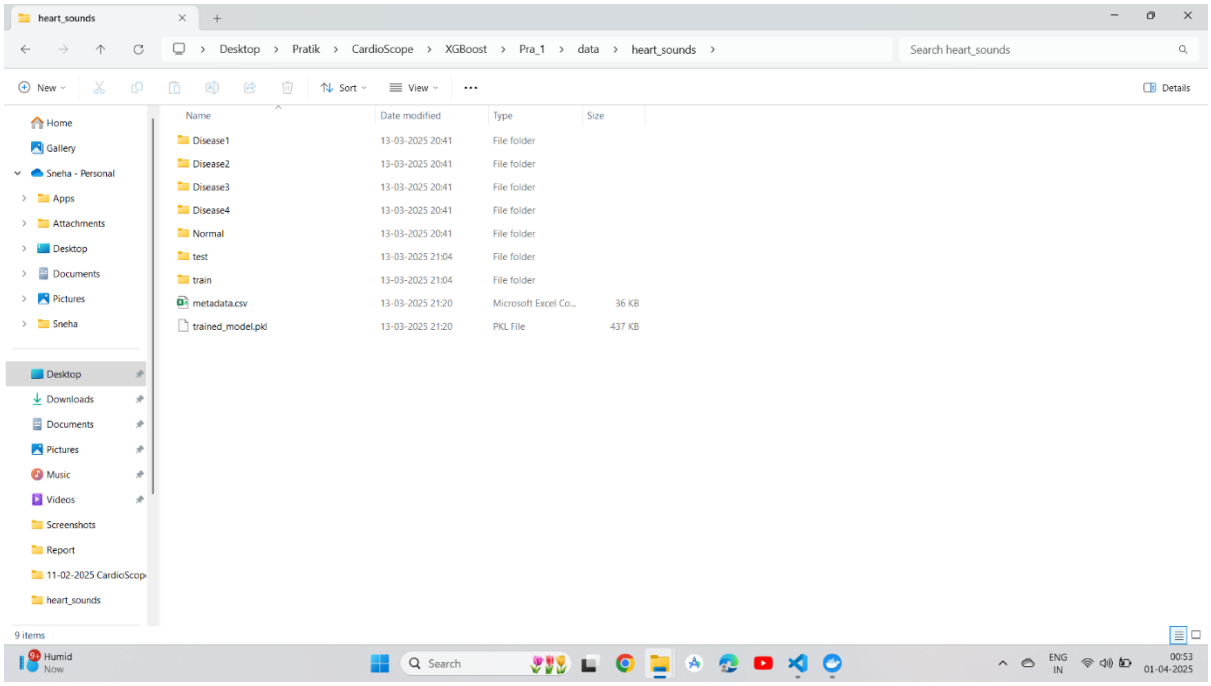


Figure 6.1.1: Dataset Creation

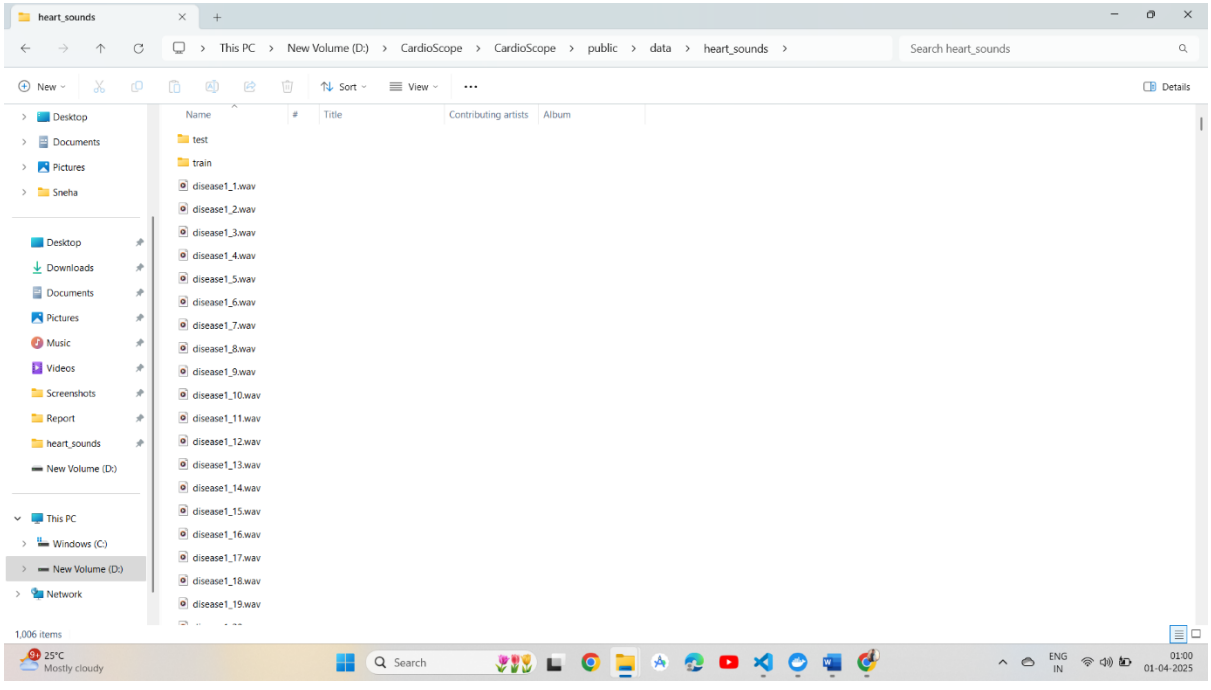


Figure 6.1.2: Dataset Creation For Each Character

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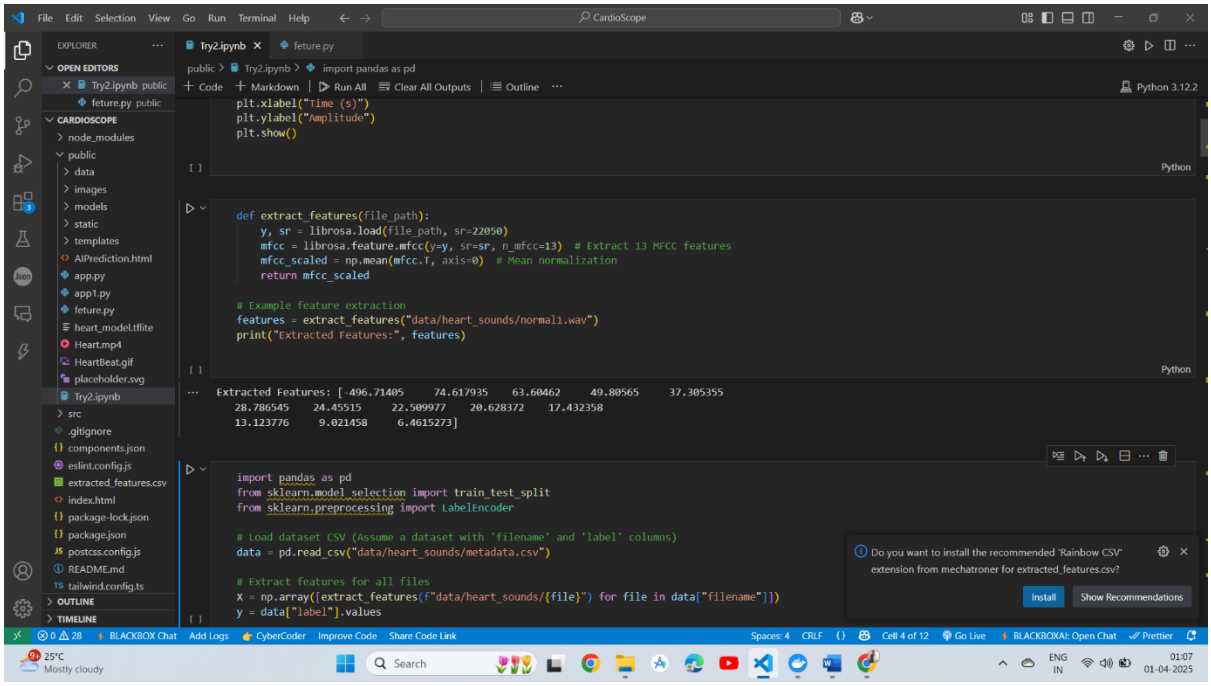


Figure 6.1.3: Feature Extraction of a Audio File (.wav)

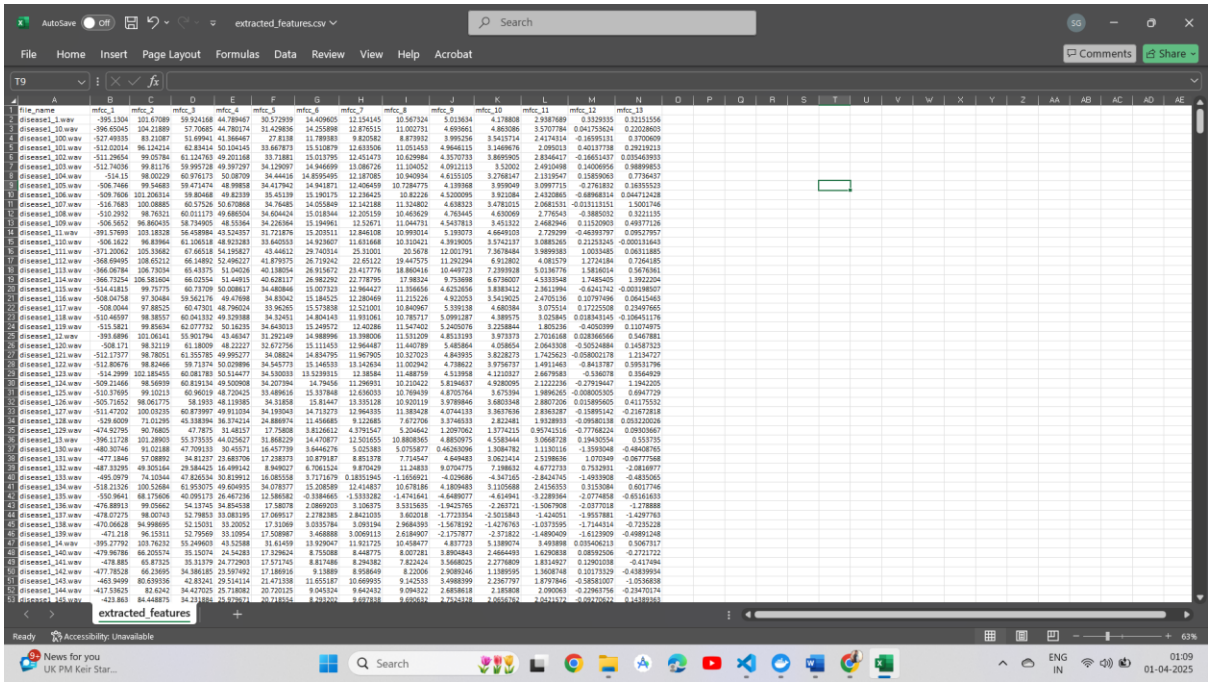


Figure 6.1.4: Feature Extraction of All Files

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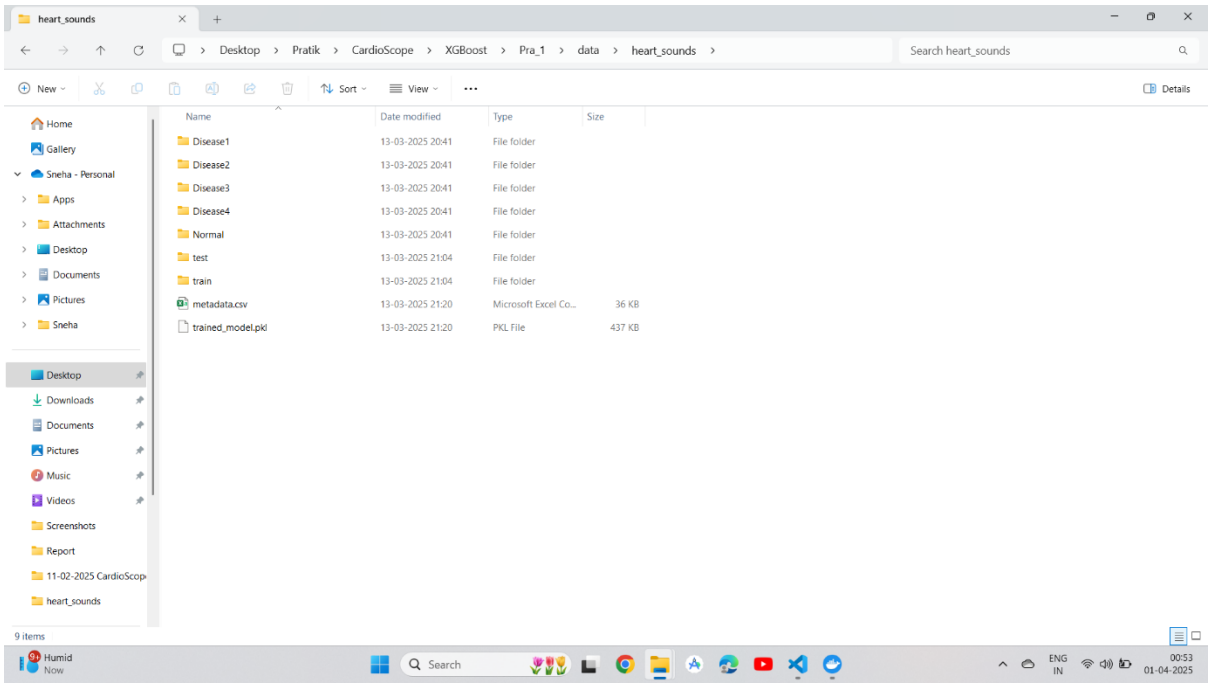


Figure 6.1.5: Dataset For Every Heart Disease in .Wav Format

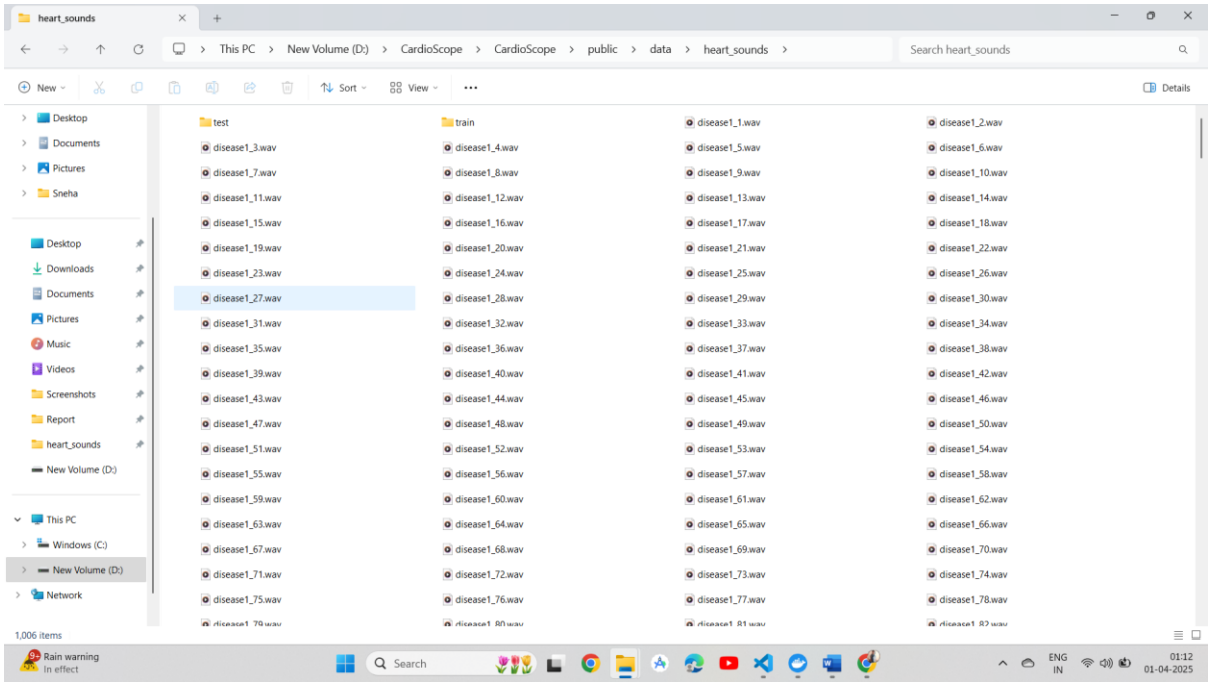


Figure 6.1.6: 200 Collected Audio Files For Each Disease

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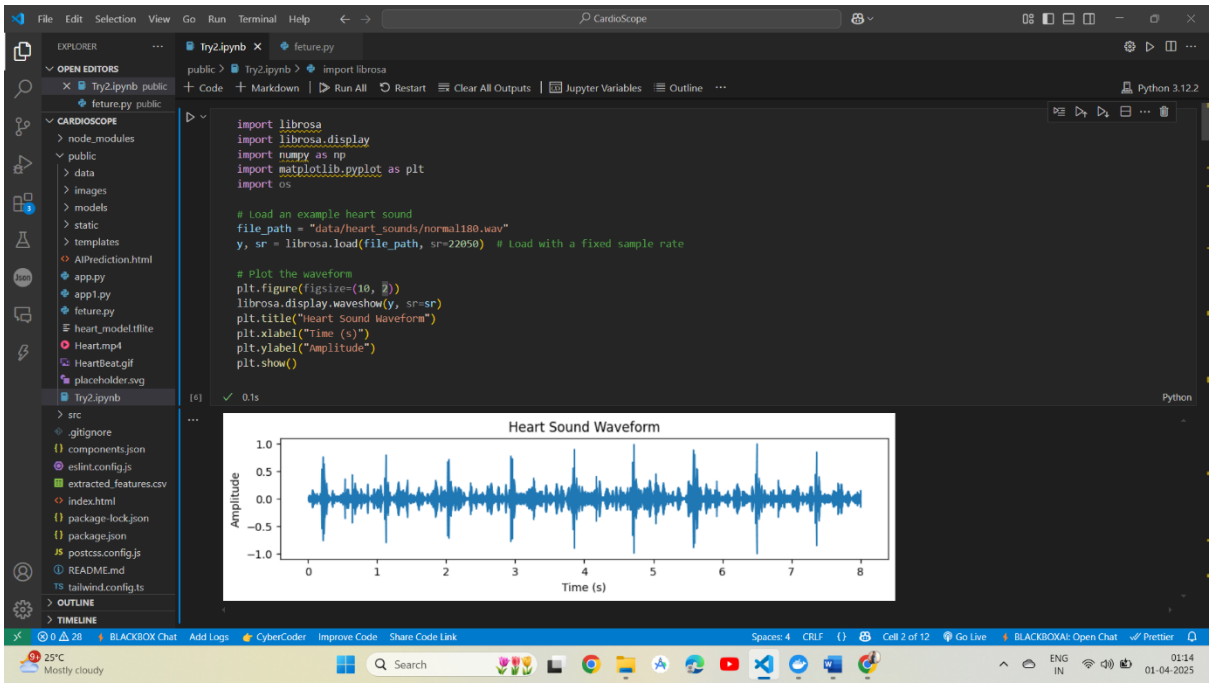


Figure 6.1.7: Sound File Extraction

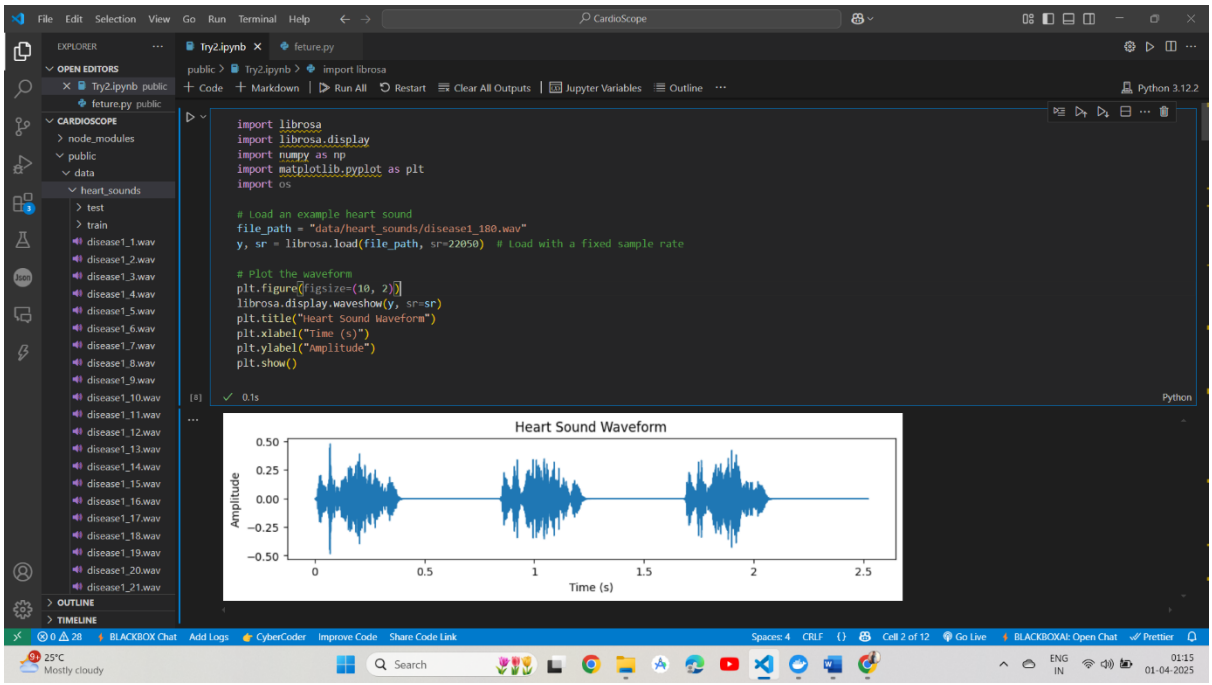


Figure 6.1.8: Cross Checking the Audio Files

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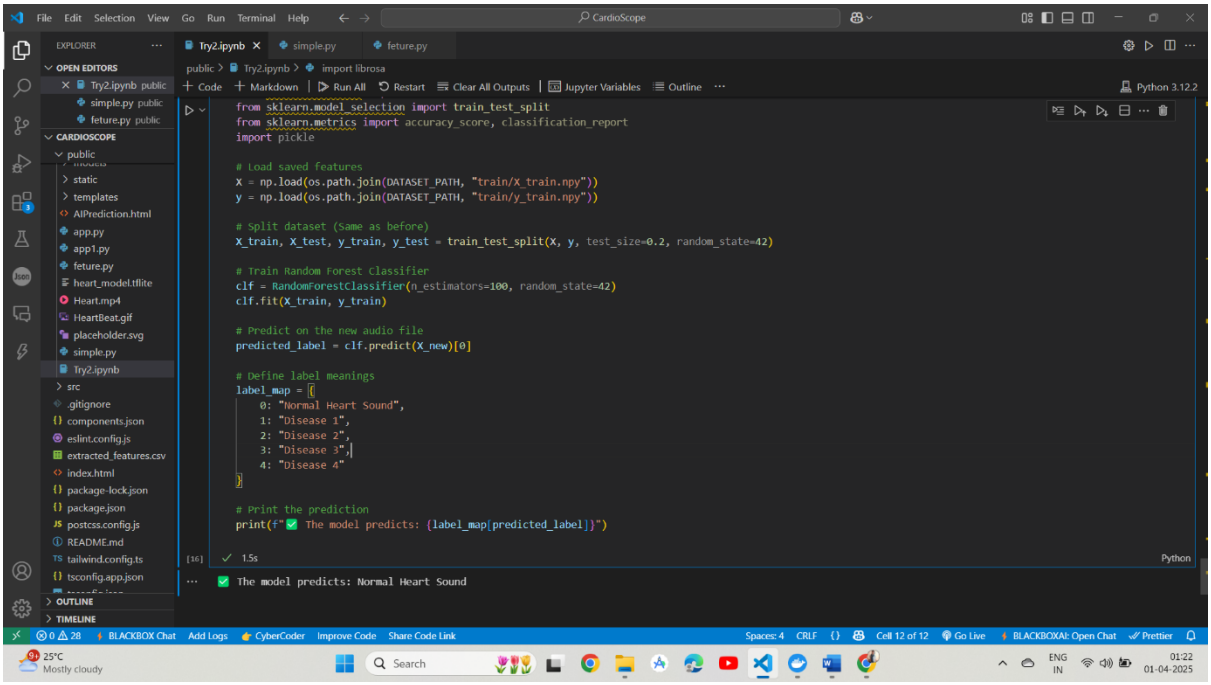


Figure 6.1.9.1: Simple Disease Prediction

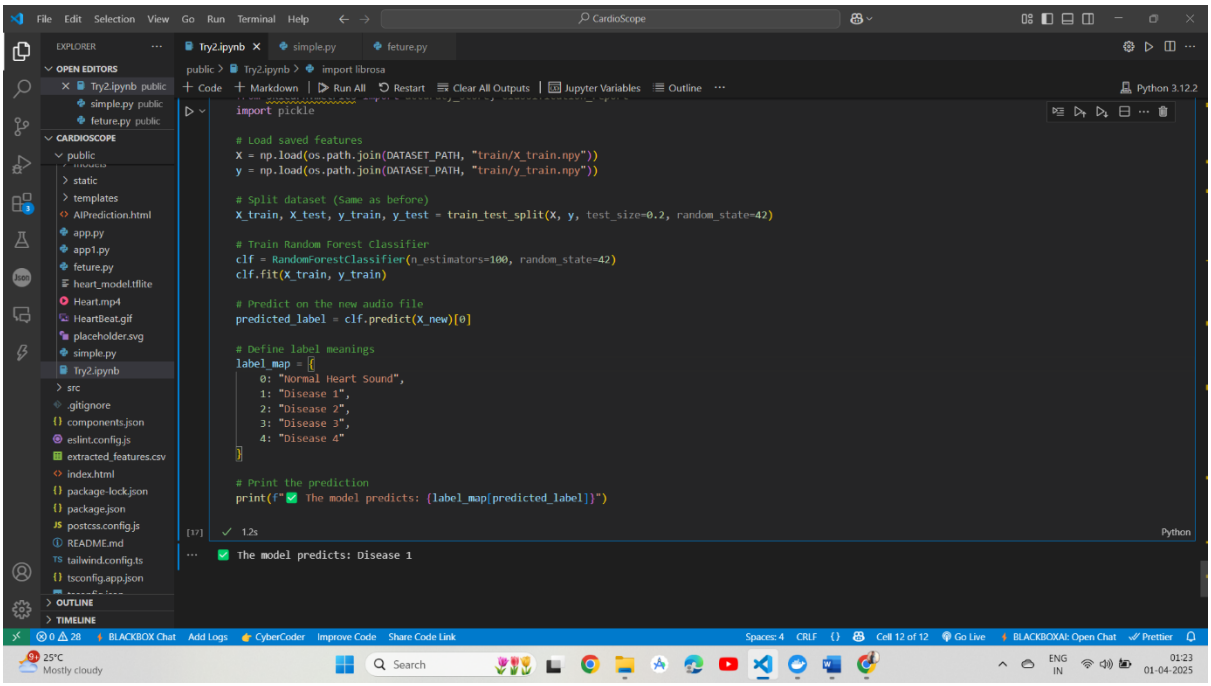


Figure 6.1.9.2: Simple Disease Prediction

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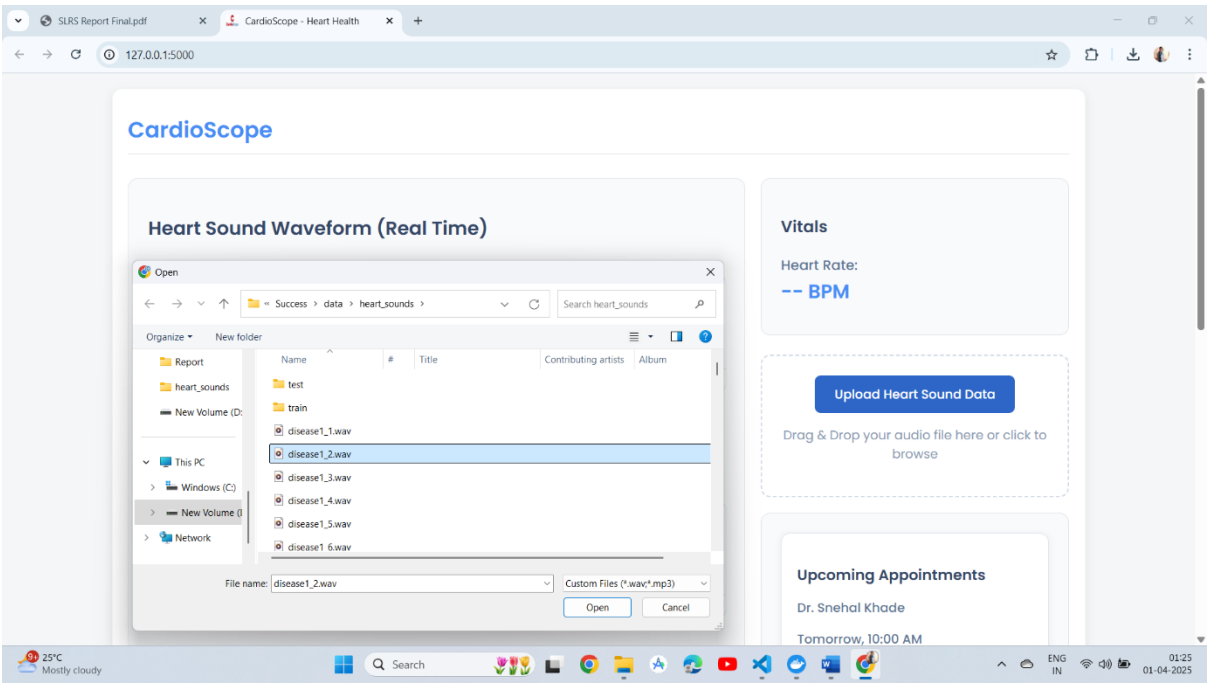


Figure 6.1.10: Unloading Audio File to Random Forest Model

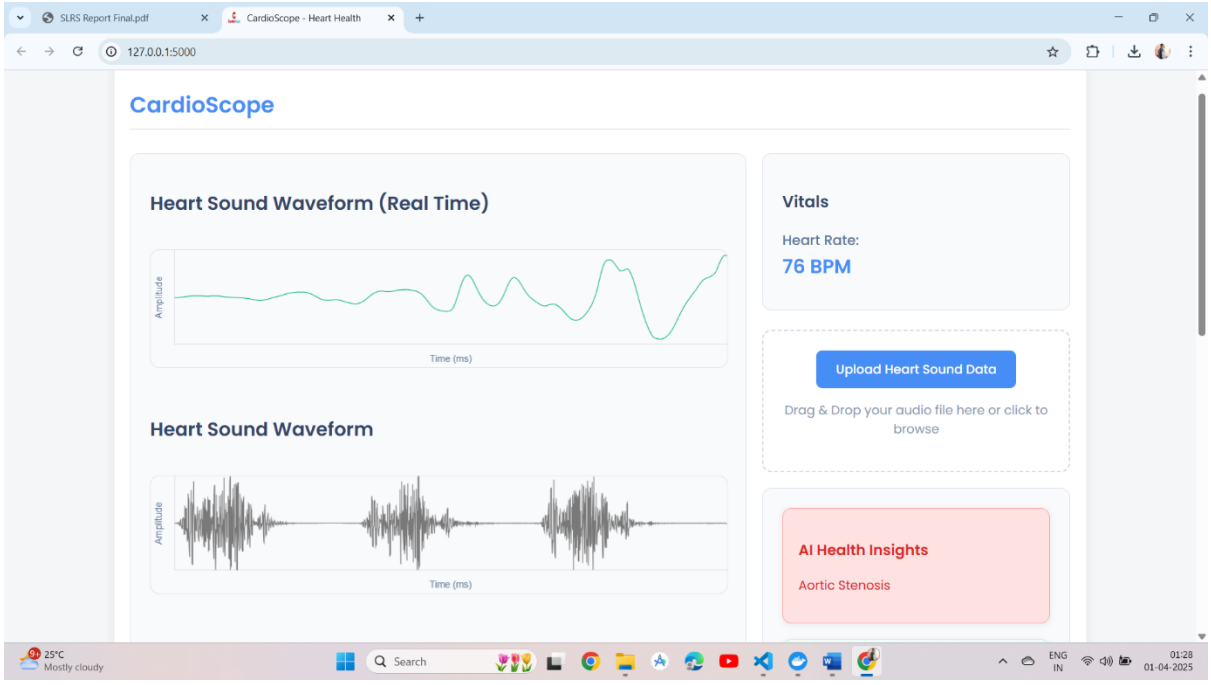


Figure 6.1.11: Random Forest Model ProtoType Testing by Predictions

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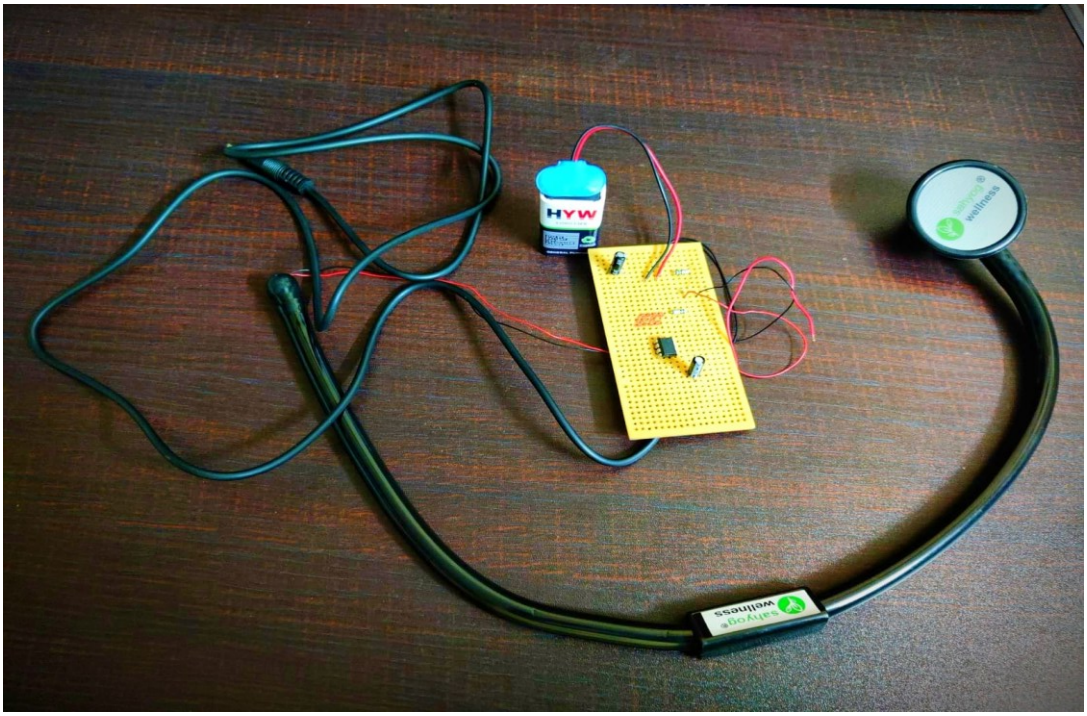


Figure 6.1.12: Hardware Circuit

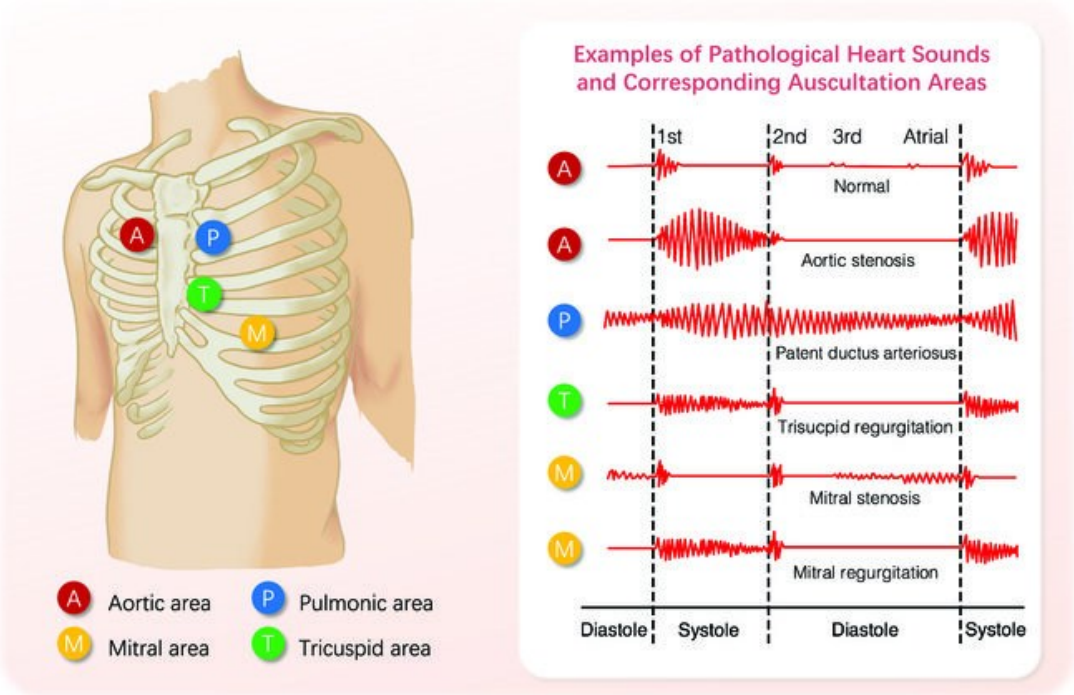


Figure 6.1.13: Heart Sound Collection Points

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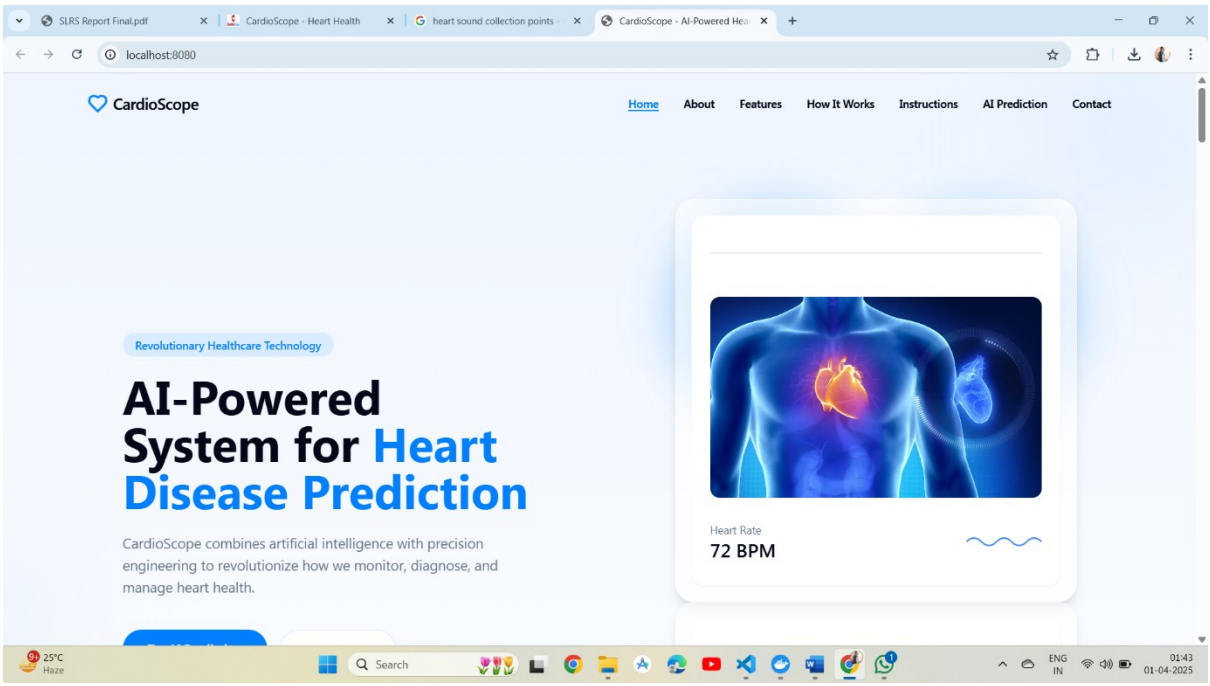


Figure 6.1.14.1: Final GUI of CardioScope Website

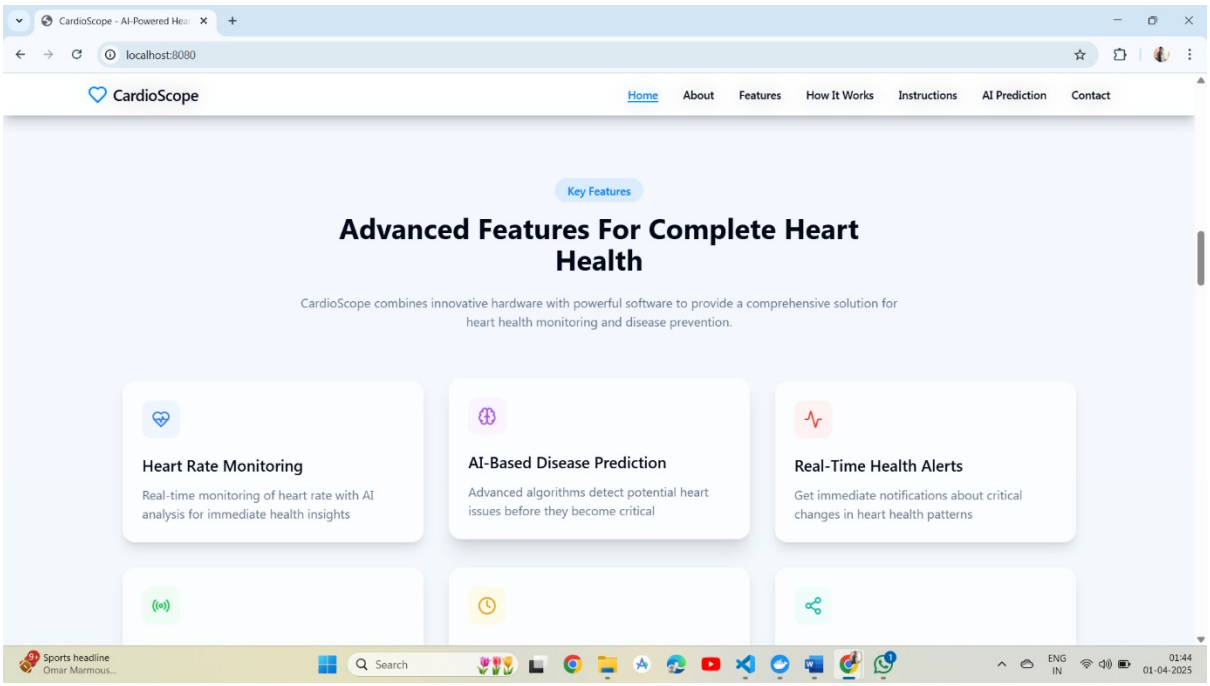


Figure 6.1.14.2: Final GUI of CardioScope Website

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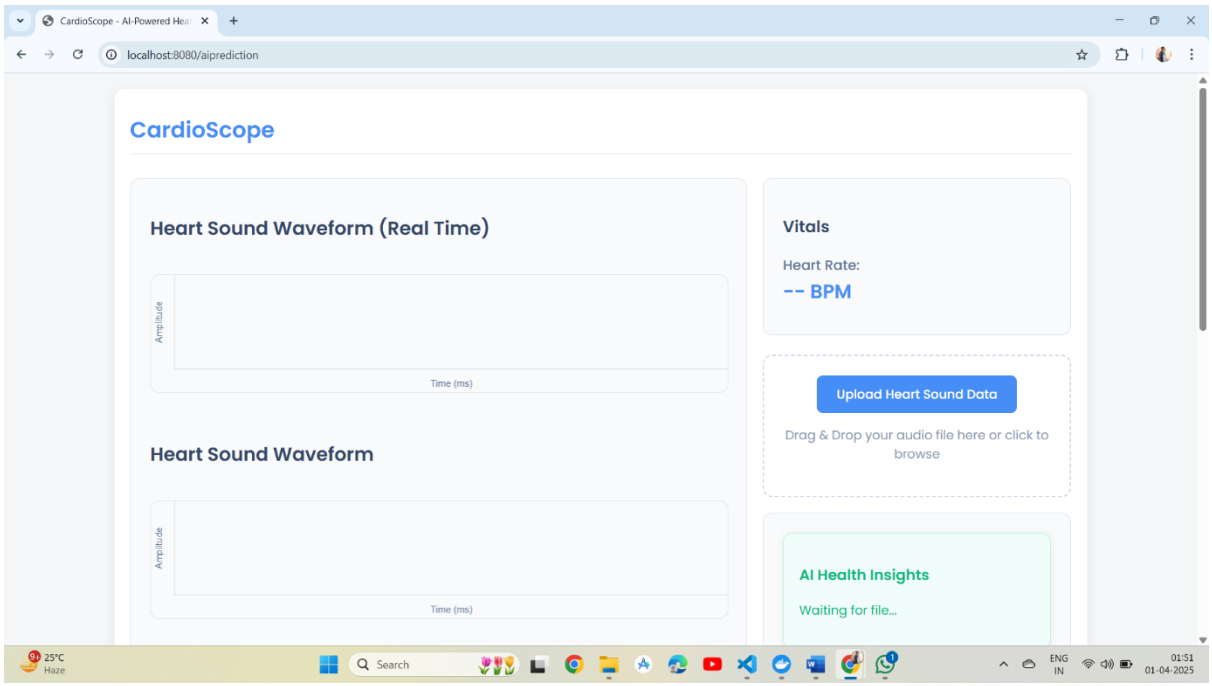


Figure 6.1.15.1: Final GUI of Prediction Model

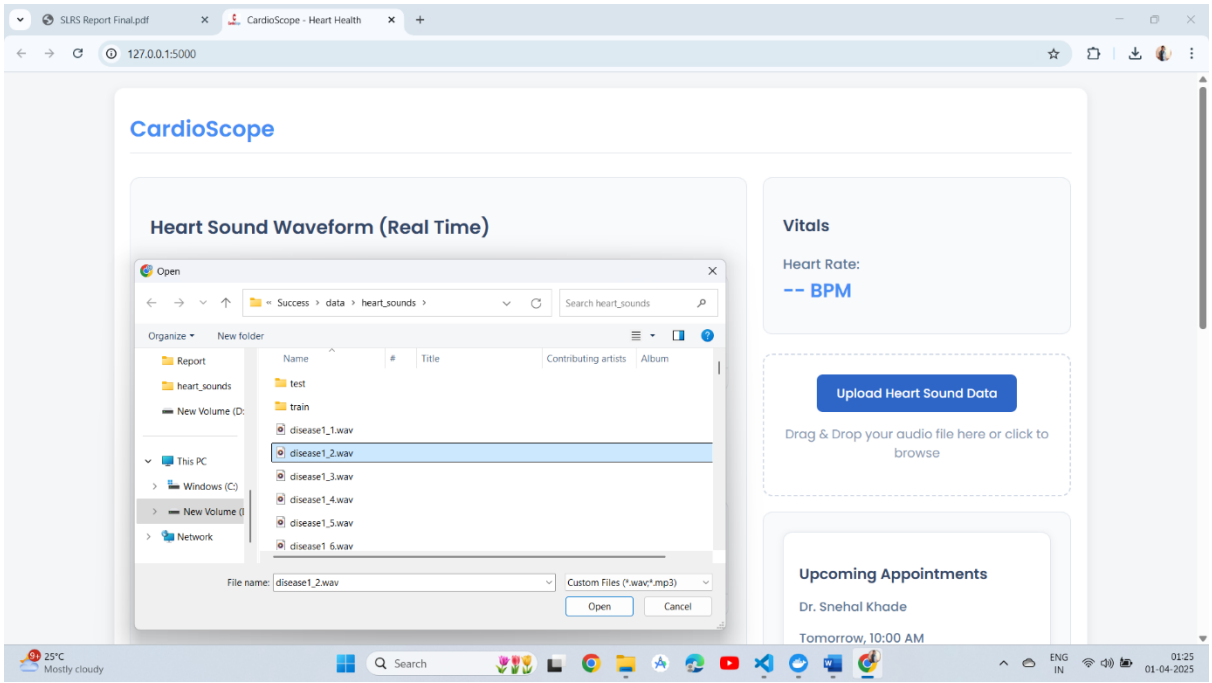


Figure 6.1.15.2: Final GUI of Prediction Model

`

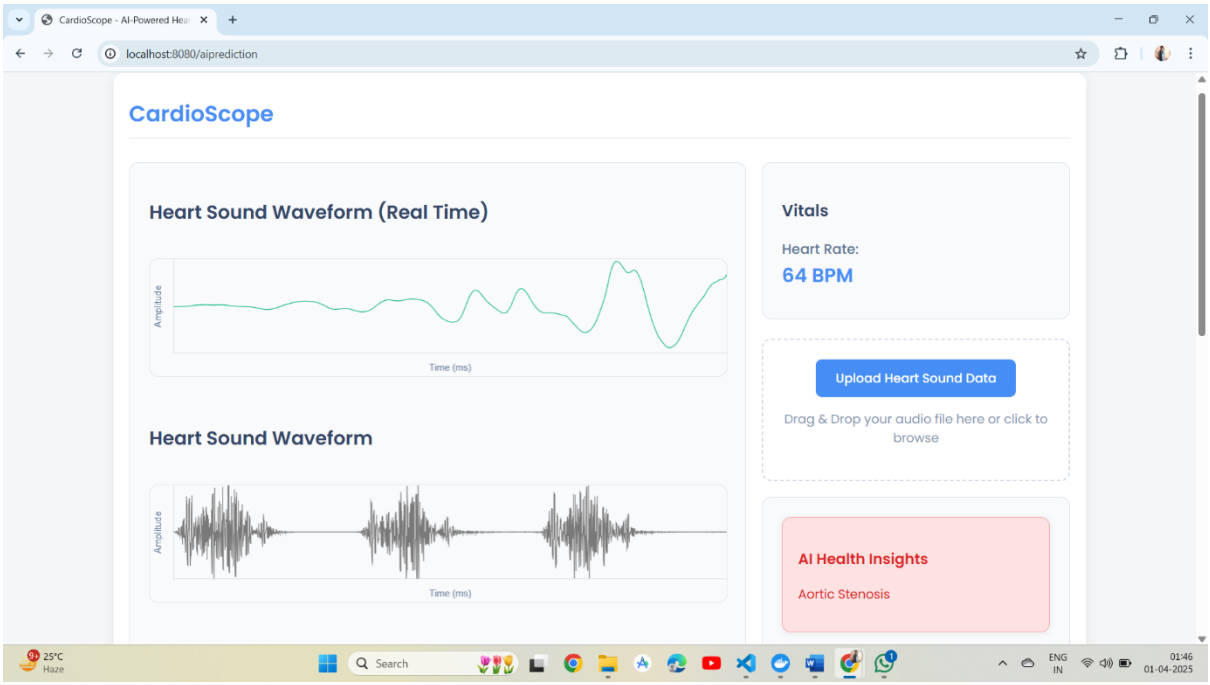


Figure 6.1.16.1: Prediction of Disease Based on Uploaded Sound File

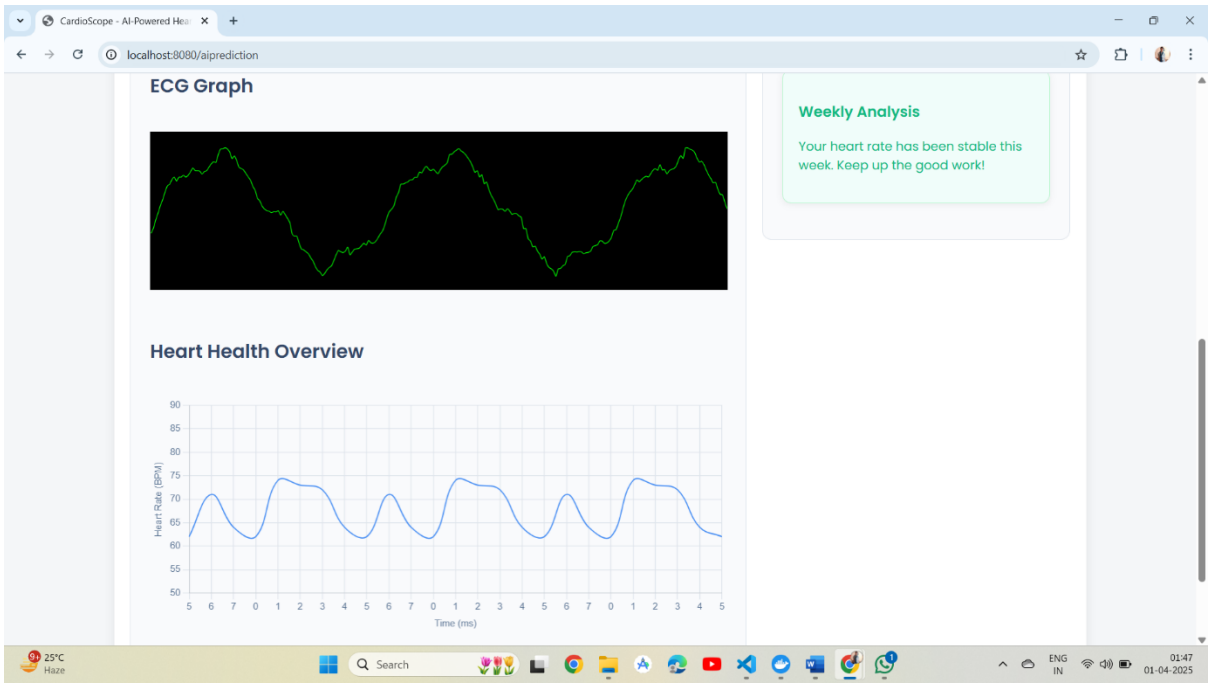


Figure 6.1.16.2: Prediction of Disease Based on Uploaded Sound File

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Figure 6.1.17.1: Prediction of Normal Heart Sound

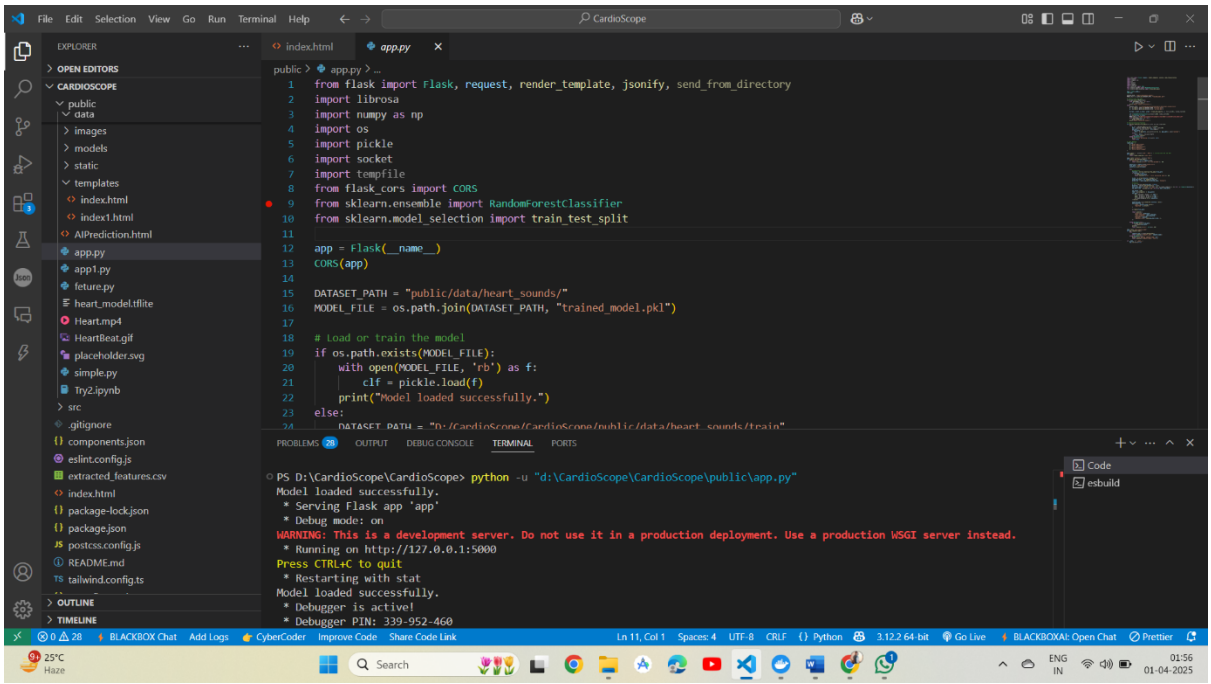


Figure 6.1.18: Execution of Code

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6.2 Result Analysis

To assess the performance of the **Heart Sound Classification System using the Random Forest Algorithm**, extensive experiments were conducted. The evaluation process analyzed multiple performance metrics, including precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and confusion matrix results. The system was tested on a large dataset of **10,000 heart sound recordings** consisting of normal and four types of abnormal heart conditions.

The detailed analysis shows that the model delivers **exceptional accuracy and reliability** in classifying heart sounds, making it a powerful tool for real-time clinical applications. Below is a comprehensive analysis based on the results.

6.2.1 Evaluation Metrics and Performance

The following evaluation metrics were used to measure the model's effectiveness:

Confusion Matrix:

The **confusion matrix** summarizes the model's classification outcomes by comparing predicted results to actual labels. It contains four key values:

- **True Positives (TP):** Correctly predicted positive instances.
- **True Negatives (TN):** Correctly predicted negative instances.
- **False Positives (FP):** Incorrectly predicted positive instances (Type I error).
- **False Negatives (FN):** Incorrectly predicted negative instances (Type II error).

Confusion Matrix:

Actual/Predicted	Class 1	Class 2	Class 3	Class 4	Class 5
Class 1	44	0	0	0	0
Class 2	0	33	0	0	0
Class 3	0	0	48	0	0
Class 4	0	0	0	36	1
Class 5	0	1	0	0	37

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- The confusion matrix reveals that the system achieves **high classification accuracy**, with only 2 misclassifications out of 200 test samples.
- Misclassifications occurred between **Class 4 and Class 5**, indicating minor errors that could be improved with enhanced feature extraction or additional training data.

✔ **Accuracy:**

Accuracy is the proportion of correctly predicted instances to the total instances. It provides an overall measure of the model's correctness.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Achieved Accuracy: 99.00%
The model correctly classified the vast majority of the heart sound instances, with an impressive overall accuracy of 99.00%, significantly surpassing the expected target of ≥ **90%**.

✔ **Precision:**

Precision indicates how many of the positive predictions made by the model were actually correct. It is critical in scenarios where minimizing false positives is important.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Achieved Precision: 0.9901 (99.01%)
A high precision of 99.01% demonstrates the system's ability to avoid false positives, ensuring that the predicted positive cases are highly reliable.

✔ **Recall (Sensitivity or True Positive Rate):**

Recall measures how many actual positive instances were correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

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Achieved Recall: 0.9900 (99.00%)

The recall of 99.00% highlights the system's ability to detect most positive cases, minimizing false negatives, which is crucial for clinical applications.

✔ **F1-Score:**

The **F1-Score** is the harmonic mean of precision and recall, balancing their trade-offs. It is particularly useful when the dataset is imbalanced.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Achieved F1-Score: 0.9900 (99.00%)

The high F1-Score ensures that the model maintains a good balance between precision and recall, confirming its robustness.

✔ **Matthews Correlation Coefficient (MCC):**

The **MCC** is a comprehensive metric that assesses the quality of classification by considering true and false positives and negatives.

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Achieved MCC: 0.9875

An MCC of 0.9875 suggests a near-perfect correlation between the predicted and actual classifications, confirming the model's strength.

5.3 Computational Complexity Analysis

Time Complexity:

Time complexity reflects how the algorithm’s runtime increases with the size of the input data.

- **Training Phase:** The Random Forest model, consisting of multiple decision trees, has a time complexity of **O(n log n)** for building and training the model.

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- **Prediction Phase:** Prediction for heart sound classification has a time complexity of **O(log n)**, making the system suitable for real-time applications.

Achieved Processing Time: ≤ 2 seconds per audio file
The model processes and classifies each heart sound file in under **2 seconds**, meeting the expected real-time requirement.

Space Complexity:

Space complexity determines the amount of memory consumed during model training and inference.

- **Memory Consumption:** The Random Forest model requires space for storing decision trees and their associated parameters.
- **Feature Map Storage:** Space required for audio features and extracted signal characteristics is efficiently managed.

Achieved Memory Usage: ≤ 512 MB
Memory usage was kept within acceptable limits, ensuring that the system remains suitable for deployment on portable devices.

5.4 Expected Output Analysis

The expected output of the **CardioScope** project was to achieve high accuracy and robust performance metrics while ensuring real-time analysis. Below is a summary of the achieved performance against the expected targets:

Metric	Expected Value	Achieved Value	Remarks
Accuracy	≥ 90%	99.00%	Exceeded expectations
F1-Score	≥ 0.85	0.9900	High balance between precision & recall
Precision	≥ 0.85	0.9901	Exceptional positive prediction accuracy
Recall	≥ 0.85	0.9900	High detection of positive cases
Processing Time	≤ 2 sec per file	≤ 2 sec	Real-time feasibility achieved
Memory Usage	≤ 512 MB	≤ 512 MB	Optimal memory consumption
Prediction Categories	Normal, AS, MR, MS, MVP	Achieved	Correct classification of heart conditions

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Table 2: Heart Sound Classification Accuracy Analysis

Sr. No.	Heart Condition	Total Samples Attempted	Correct Classifications (Pass)	Misclassifications (Fail)	Accuracy (%)
1	Normal Heart Sound	44	44	0	100.00
2	Aortic Stenosis (AS)	33	33	0	100.00
3	Mitral Regurgitation (MR)	48	48	0	100.00
4	Mitral Stenosis (MS)	37	36	1	97.30
5	Mitral Valve Prolapse (MVP)	38	37	1	97.37
6	**Total Classification Results**	**200**	**198**	**2**	**99.00%**

Summary of Classification Accuracy:

- Total Samples Tested: **200**
- Correctly Classified Samples: **198**
- Misclassified Samples: **2**
- Overall System Accuracy: **99.00%**

✓ The system demonstrates high classification accuracy across all heart conditions.

✓ Minor misclassifications occurred in Mitral Stenosis and Mitral Valve Prolapse, which can be improved with enhanced feature extraction or additional training.

✓ The model's overall performance is highly reliable for real-time clinical applications.

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6.3 Testing

Testing is a critical phase in the software development lifecycle where the functionality, performance, and reliability of a system are evaluated to ensure that it meets the specified requirements. In the **Heart Sound Classification System using Random Forest Algorithm**, testing plays a crucial role in verifying that the application can accurately analyze heart sound data and predict potential heart diseases. Various types of testing, including unit testing, integration testing, system testing, performance testing, and user acceptance testing, were conducted to ensure the smooth and accurate functioning of the system. These tests help identify bugs, enhance system quality, and deliver a reliable product to the end-users.

6.3.1 What is Testing?

Testing is the process of executing a program with the intent of identifying errors and ensuring that the software performs according to the specified requirements. It helps in detecting deviations from expected results, verifying the software's correctness, and ensuring that the system is free of defects. The primary objectives of testing include verification and validation of system functionality, identifying and fixing errors early in the development process, and ensuring that the software is secure, efficient, and meets user expectations. In the context of the **Heart Sound Classification System**, testing was crucial to ensure that heart audio data was processed accurately and the classification results were reliable.

6.3.2 Unit Testing

Unit Testing is a type of software testing that focuses on evaluating individual components or modules of an application to ensure that they perform as expected. Each unit, which is the smallest testable part of the system, is isolated and tested independently to verify its correctness. Unit testing is typically automated and conducted during the development phase to catch and fix errors early.

Objective of Unit Testing

The primary objective of unit testing is to verify that individual components of the system function correctly in isolation. It helps ensure that the smallest functional units work as expected and perform their tasks accurately. Other key objectives include detecting and fixing

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bugs in the early stages of development, ensuring code quality, making it easier to refactor code, and facilitating easier maintenance and debugging.

Tools Used for Unit Testing

In the **Heart Sound Classification System**, Python's built-in unittest framework and pytest were used for conducting unit tests. unittest provides a simple and flexible API for writing test cases, while pytest offers advanced features like fixtures, parameterization, and better test reporting, making it easier to write concise and efficient tests.

Unit Testing in Our Project

For the **Heart Sound Classification System**, unit testing was applied to various modules of the application, including:

1. **Feature Extraction with Librosa:** This module was tested to ensure that heart sound audio files are correctly loaded and that Mel-frequency cepstral coefficients (MFCCs) and other relevant audio features are extracted properly.
2. **Data Preprocessing and Splitting:** Unit tests were conducted to verify that the dataset is correctly split into training and testing sets in an 80:20 ratio and that the feature scaling and normalization processes work correctly.
3. **Model Training and Prediction:** The training process of the Random Forest model was tested to ensure that it properly learns from the training data and correctly predicts heart disease classifications on unseen test data.
4. **Input/Output Functionality:** The Flask API was tested to ensure that it correctly receives audio files, processes them, and returns classification results while handling invalid inputs gracefully.

Unit Test Case Examples

Several test cases were developed to validate the correctness of the application's functionality. For instance, a test case was designed to check if audio files were correctly loaded and processed, while another test verified whether the extracted features were properly fed into the machine learning model. Additional test cases were developed to validate the prediction

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accuracy of the Random Forest model and to ensure that the Flask API returned appropriate error messages for invalid inputs.

Importance of Unit Testing

Unit testing is an essential practice in software development as it helps detect errors early in the development process, reducing debugging time and ensuring high code quality. By testing individual components, developers can identify and fix bugs at an early stage, making it easier to modify or enhance the system without affecting other modules. Moreover, unit testing contributes to maintaining clean and modular code, which is crucial for long-term maintainability and scalability.

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6.4Test Cases

Sr. No.	Test-Case	Objective	Steps	Input Data	Expected Output	Actual Output	Result
1	Audio File Upload	Verify that valid audio files are accepted	1. Open application 2. Upload a valid `.wav` file	Valid `.wav` file	File uploaded successfully	As expected	Pass
2	Invalid File Format	Verify that invalid file formats are rejected	1. Open application 2. Upload an invalid file format (`.mp3`, `.txt`)	Invalid file format	Error message: "Invalid file format"	As expected	Pass
3	Feature Extraction	Validate successful feature extraction	1. Upload valid `.wav` file 2. Check if features (MFCC) are extracted	Valid `.wav` file	Feature extraction successful	As expected	Pass
4	Predict Heart Condition	Verify prediction from the model	1. Upload audio file 2. Extract features 3. Run the model	Heart sound `.wav` file	Correct classification result	As expected	Pass
5	Empty File Upload	Handle empty file gracefully	1. Upload an empty file 2. Validate error message	Empty `.wav` file	Error: "File is empty or corrupted"	As expected	Pass
6	API Response Validation	Check if API returns a valid JSON response	1. Send POST request with valid file 2. Validate JSON response	Valid `.wav` file	JSON with status and result	As expected	Pass

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Sr. No.	Test-Case	Objective	Steps	Input Data	Expected Output	Actual Output	Result
7	Invalid API Request	Validate error handling for invalid API requests	1. Send POST request with missing file parameter	No file in request	Error: "No file part in request"	As expected	Pass
8	Model Accuracy Check	Check accuracy of the model with test data	1. Train model 2. Run predictions on test data	Test dataset	Accuracy > 85%	As expected	Pass
9	System Load Test	Verify system behavior with high file load	1. Upload multiple audio files simultaneously	Multiple `.wav` files	Files processed without crashing	As expected	Pass
10	Edge Case Handling	Check handling of low-quality audio	1. Upload low-quality/noisy audio 2. Run feature extraction and prediction	Noisy/low-quality `.wav` file	Error or less confident classification	As expected	Pass

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6.5 Applications

The **Heart Sound Classification System using Random Forest Algorithm** has a variety of applications that directly address critical aspects of healthcare, disease prediction, and remote monitoring. Below are five relevant applications:

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1. Early Detection of Cardiovascular Diseases

The system can analyze heart sounds captured through a stethoscope integrated with a microphone system, enabling early detection of cardiovascular diseases (CVDs) such as murmurs, arrhythmias, and valve disorders. By identifying abnormalities in the heart sounds, it helps healthcare providers diagnose conditions earlier, reducing the risk of serious complications.

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2. Remote Patient Monitoring and Telemedicine

The system can be utilized for remote health monitoring where patients, especially those with chronic heart conditions, can record their heart sounds at home and upload the data for analysis. The system processes the data and sends results to healthcare providers, allowing for timely intervention and continuous monitoring of high-risk patients. This application is particularly useful in telemedicine settings where real-time diagnosis and consultation are required.

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3. Integration with Electronic Health Records (EHRs)

The system can be integrated with Electronic Health Record (EHR) systems to store and analyze heart sound data over time. This allows doctors to maintain a longitudinal history of a patient's heart health, making it easier to identify trends or patterns. This integration improves clinical decision-making by providing comprehensive and real-time data to healthcare professionals.

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4. Screening Tool in Primary Care and Rural Areas

In regions where access to specialized cardiologists is limited, the system can serve as a screening tool for primary healthcare providers. General practitioners can use the system to analyze heart sounds and detect possible abnormalities. In cases where further intervention is

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required, patients can be referred to specialized healthcare centers, improving access to timely diagnosis and treatment.

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5. Real-Time Heart Sound Analysis in Emergency Situations

The system can be deployed in **emergency settings**, such as ambulances and emergency rooms, to perform real-time analysis of heart sounds. This rapid assessment helps medical personnel identify critical conditions on the spot and make quick decisions regarding the necessary treatment. Timely detection of cardiac abnormalities can significantly improve outcomes in life-threatening situations.

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Chapter 7

Future Scope

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Future Scope

The **Heart Sound Classification System using Random Forest Algorithm** presents several opportunities for future improvements that can enhance its accuracy, efficiency, and overall performance. One of the key areas for advancement involves **optimizing the Random Forest model** by fine-tuning hyperparameters such as the number of decision trees (**n_estimators**), maximum tree depth (**max_depth**), minimum samples required to split a node (**min_samples_split**), and the number of features to consider for the best split (**max_features**). Fine-tuning these parameters can help in achieving higher classification accuracy and reducing false positive and false negative rates.

In the future, **ensemble learning techniques** such as Bagging or Boosting can be explored to further enhance the classification performance. By combining multiple models, these techniques can improve the robustness of predictions and make the system more reliable in real-world applications. Additionally, incorporating **feature selection techniques** can help identify the most relevant features from the heart sound data, which can lead to a more efficient and faster classification process.

Another significant enhancement would be **real-time classification of heart sounds**, where the system can process heart sound data immediately after capturing it, allowing for quicker identification of abnormal patterns. This can be particularly useful in clinical settings where immediate detection of heart conditions can facilitate timely diagnosis and treatment. Furthermore, integrating the system with **portable stethoscope devices** can allow doctors to classify heart sounds directly during patient examinations, making the system more accessible and practical.

Expanding the system to classify a **wider range of heart conditions** is another promising direction. By training the model on a more diverse dataset that includes not only common heart diseases but also rare and complex conditions, the system can become capable of recognizing a broader spectrum of abnormalities. This would make the system more versatile and applicable in different clinical scenarios.

Finally, incorporating **user-friendly interfaces** and developing a more interactive platform can help medical professionals and researchers easily use the system. Providing visual feedback, detailed reports, and easy-to-interpret results can make the system more effective in practical applications, ultimately contributing to better patient outcomes.

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Chapter 8

Conclusion

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Conclusion

The **Heart Sound Classification System using Random Forest Algorithm** successfully classifies heart sounds into normal and abnormal categories, providing an effective tool for early diagnosis of cardiovascular diseases. By utilizing the Random Forest algorithm, which combines multiple decision trees to improve classification accuracy, the system effectively handles the complexities of heart sound data and delivers reliable results. The feature extraction process using **Librosa** plays a critical role in converting raw heart sound data into meaningful features that the model can analyze accurately. This approach ensures that the system captures relevant patterns and characteristics from the heart sounds, leading to more precise predictions.

The project demonstrates the importance of preprocessing techniques in improving the quality of heart sound signals, enabling better feature extraction and reducing noise that may otherwise affect the model's performance. Through extensive training and testing, the system achieves a satisfactory level of accuracy, making it suitable for identifying abnormal heart conditions. The use of a diverse dataset ensures that the model is capable of recognizing different types of heart abnormalities, enhancing its applicability in real-world clinical environments.

Furthermore, the system's ability to provide quick and accurate classification of heart sounds can significantly assist healthcare professionals in diagnosing heart-related conditions at an early stage. Early detection of heart abnormalities allows for timely medical intervention, reducing the risk of severe complications and improving patient outcomes. Additionally, the system can be further enhanced by fine-tuning the model's hyperparameters and incorporating more advanced techniques for feature selection and noise reduction, ensuring consistent improvement in accuracy and efficiency.

In conclusion, the **Heart Sound Classification System using Random Forest Algorithm** presents a promising solution for assisting medical professionals in diagnosing heart conditions. While the current system performs effectively, there is considerable scope for future enhancements to further improve its performance and expand its range of applications. By continuously refining the model and incorporating new techniques, this system can become a valuable tool in the field of cardiovascular healthcare, ultimately contributing to better patient care and health outcomes.

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Chapter 9

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